

ECSE 4660/6660: Internetworking of Things

Lecture 13: Network Layer – Part I: Routing (DSR, AODV)

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Spring 2026

Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - TA hours 4-5 PM daily in 5312 lab
 - Lab 3 due Fri, 2/27 (BLE, Zbee)
 - Lab 4 due Fri, 3/20 (DSR, AODV, ARP)
- HW
 - HW3 due Fri, 3/13
 - No Late submission allowed
- Remote class 2/26, 3/23
- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - **Work with TA to get the hardware needed (respond on piazza)**
 - Mid-term report due 3/20 (2-3 weeks)
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

References

1. DSR

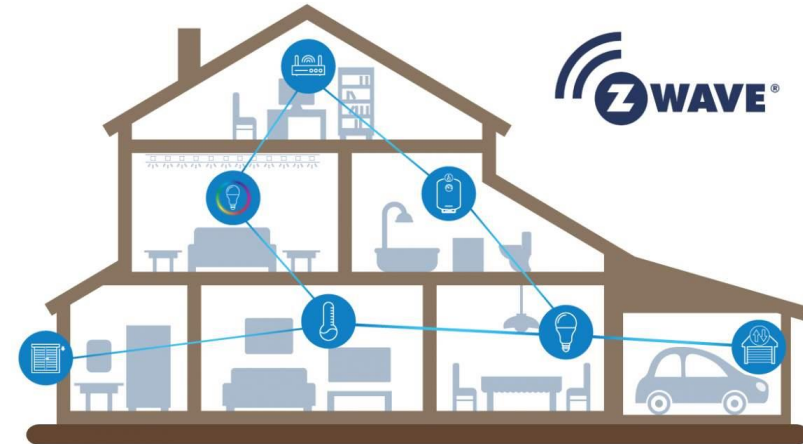
- Utilized in IoT standards (e.g. ZWave)
- References (papers, available online)
- David B. Johnson and David A. Maltz. “Dynamic Source Routing in Ad Hoc Wireless [Dynamic Source Routing in Ad Hoc Wireless Networks](#),” In *Mobile Computing*, edited by Tomasz Imielinski and Hank Korth, chapter 5, pages 153–181. Kluwer Academic Publishers, 1996.

2. AODV: Routing Protocols for Ad Hoc / Mobile Networks of Devices

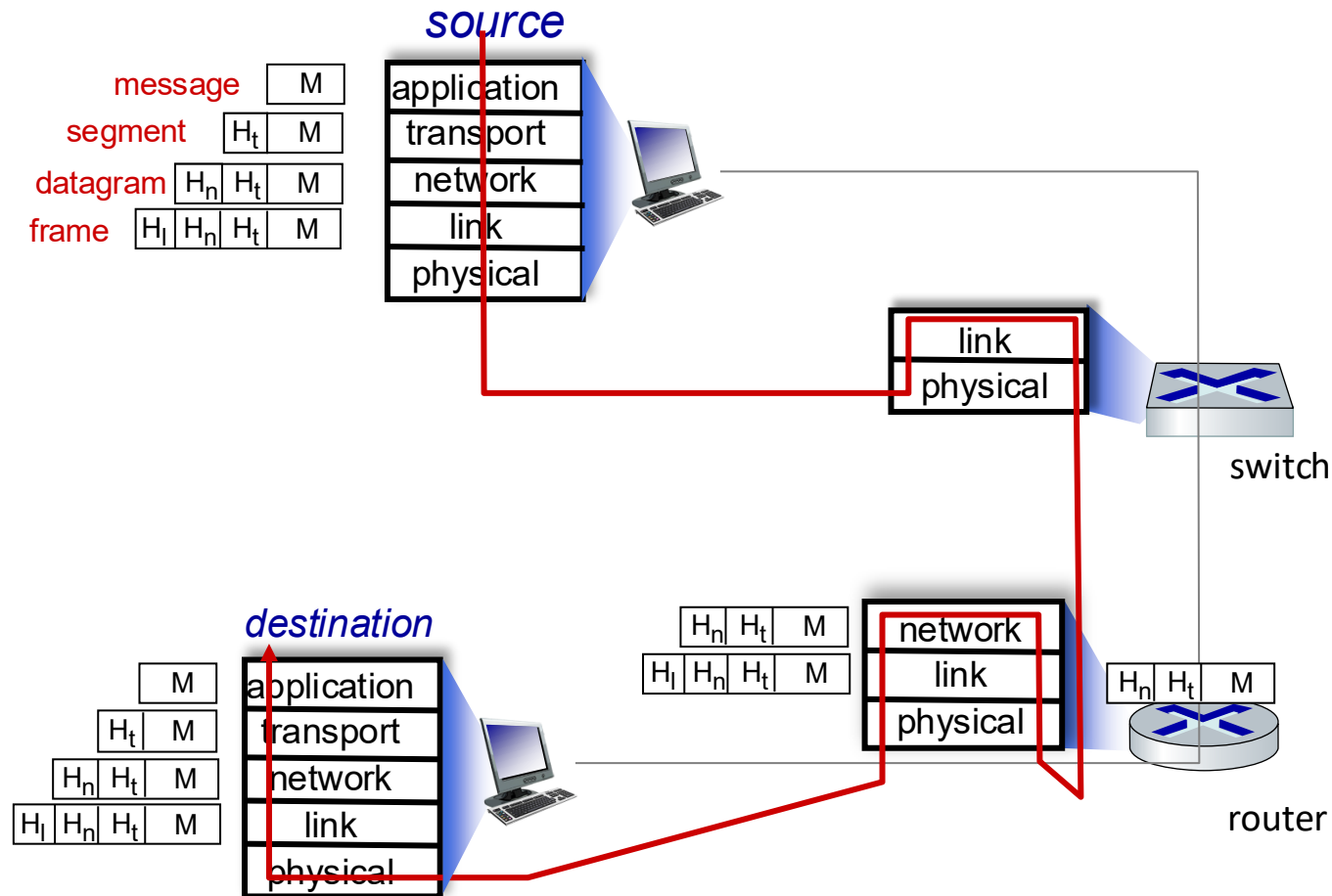
- Utilized in IoT standards (e.g. ZigBee)
 - References (papers, available online)
 - Perkins, C.E.; Royer, E.M, “[Ad-hoc on-demand distance vector routing](#),” *Mobile Computing Systems and Applications*, 1999. Proceedings. WMCSA '99. Second IEEE Workshop on , 25-26 Feb. 1999 Page(s): 90 –100.
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- Acknowledgment: DSR & AODV example slides are based in part on a conference tutorial by Prof. Nitin Vaidya, GU.

Zwave

- Z-Wave, developed by Zensys, is a two-way RF mesh networking technology that targets home control and energy management applications.
- Supports up to 232 nodes
- Source Routing
- Smart home automation



Recall: Layers in Internetworking stack

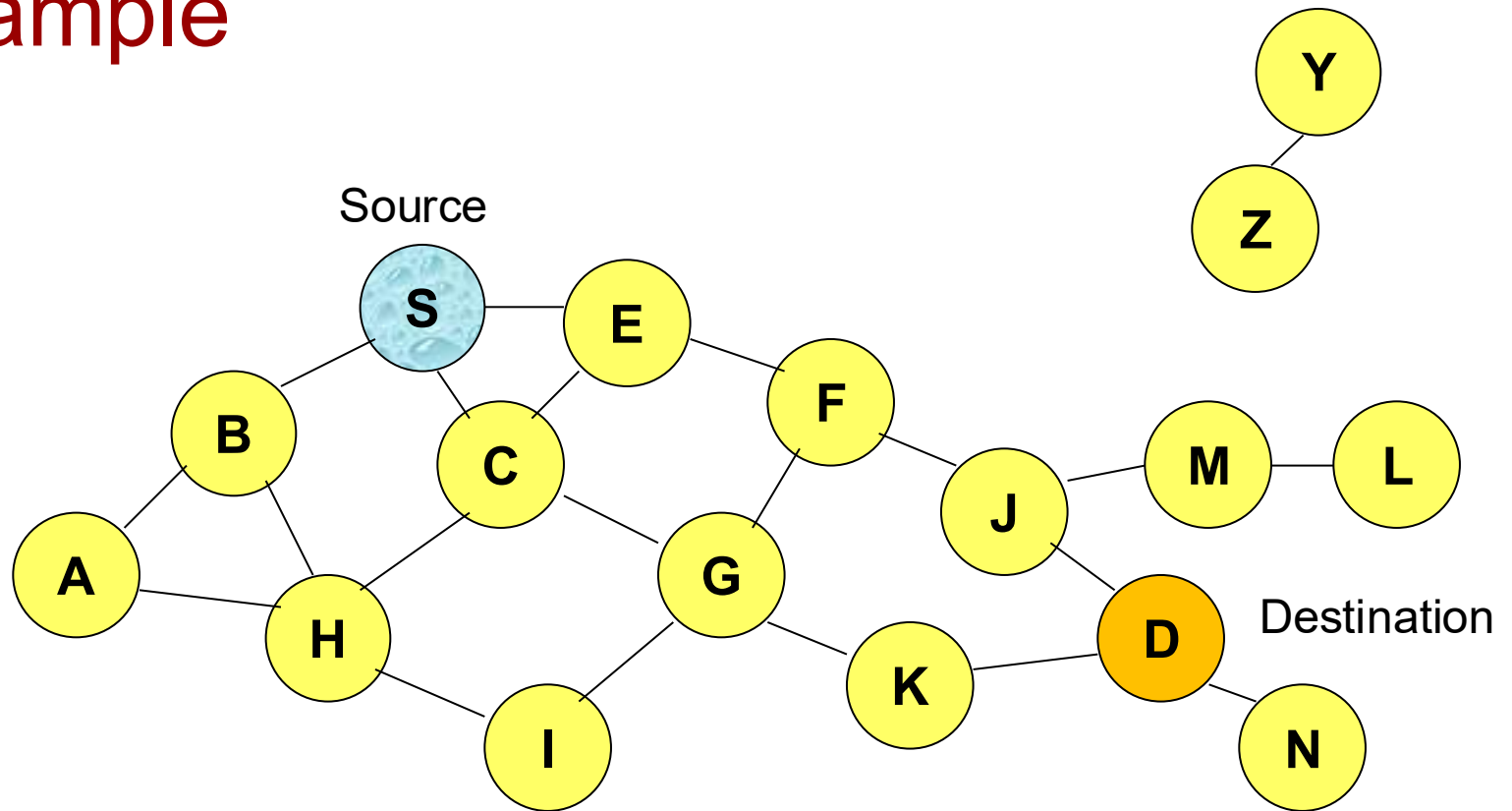


Routing: from wired to wireless

- Ad-hoc networks have dynamic time-dependent topology
 - Arcs (Edges) added/deleted
 - Nodes (Vertices) addition/deletion
 - Bi-directional or uni-directional links
 - Wireless medium is different from wired
 - It is inherently a broadcast medium
 - Fading, shadowing cause burst errors and/or intermittent connectivity
- Need to discover and update (rediscover) the network topology

Flooding

Flooding for Data Delivery-a wireless example



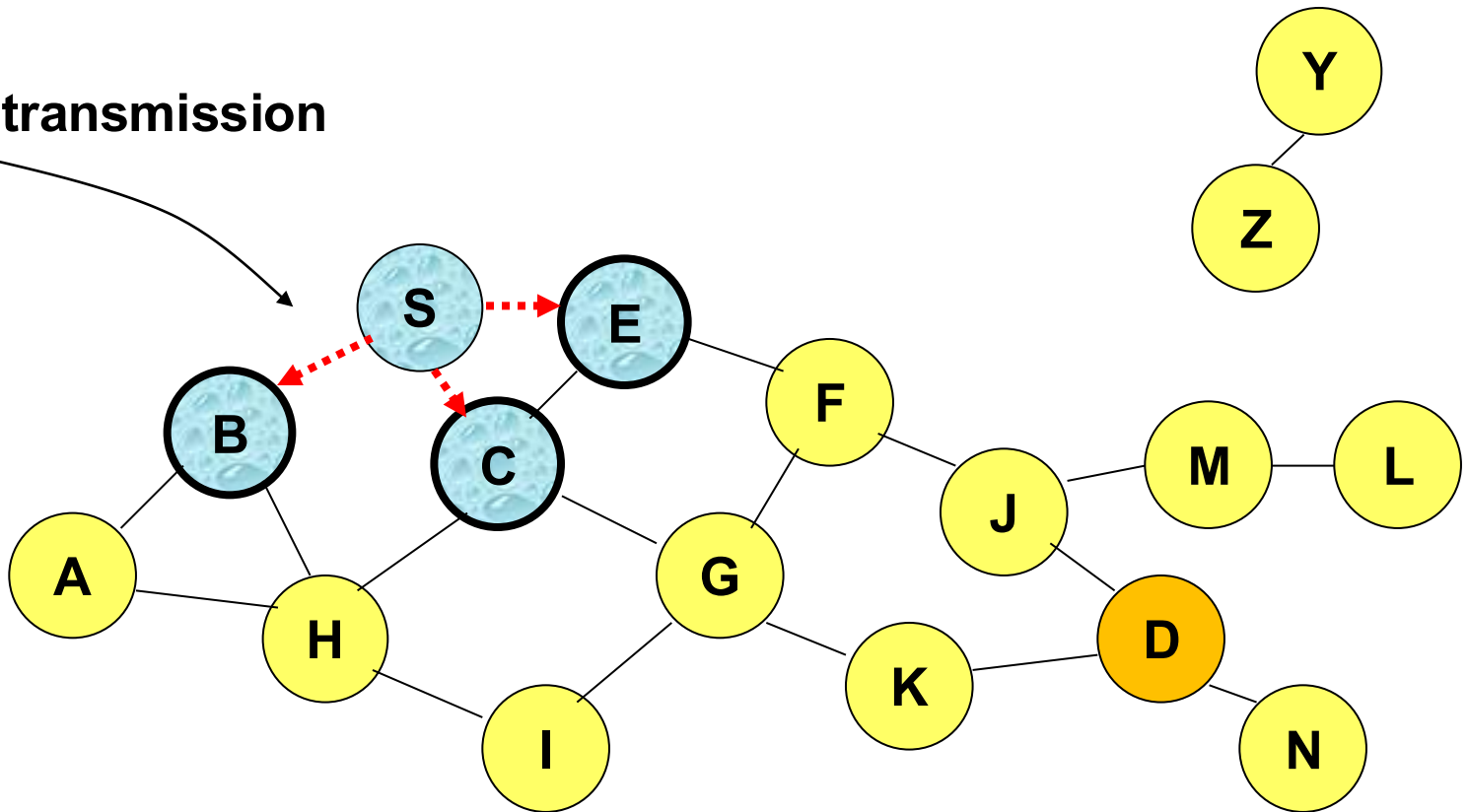
Represents a node that has received packet P



Represents that connected nodes are within each other's transmission range

Flooding for Data Delivery

Broadcast transmission

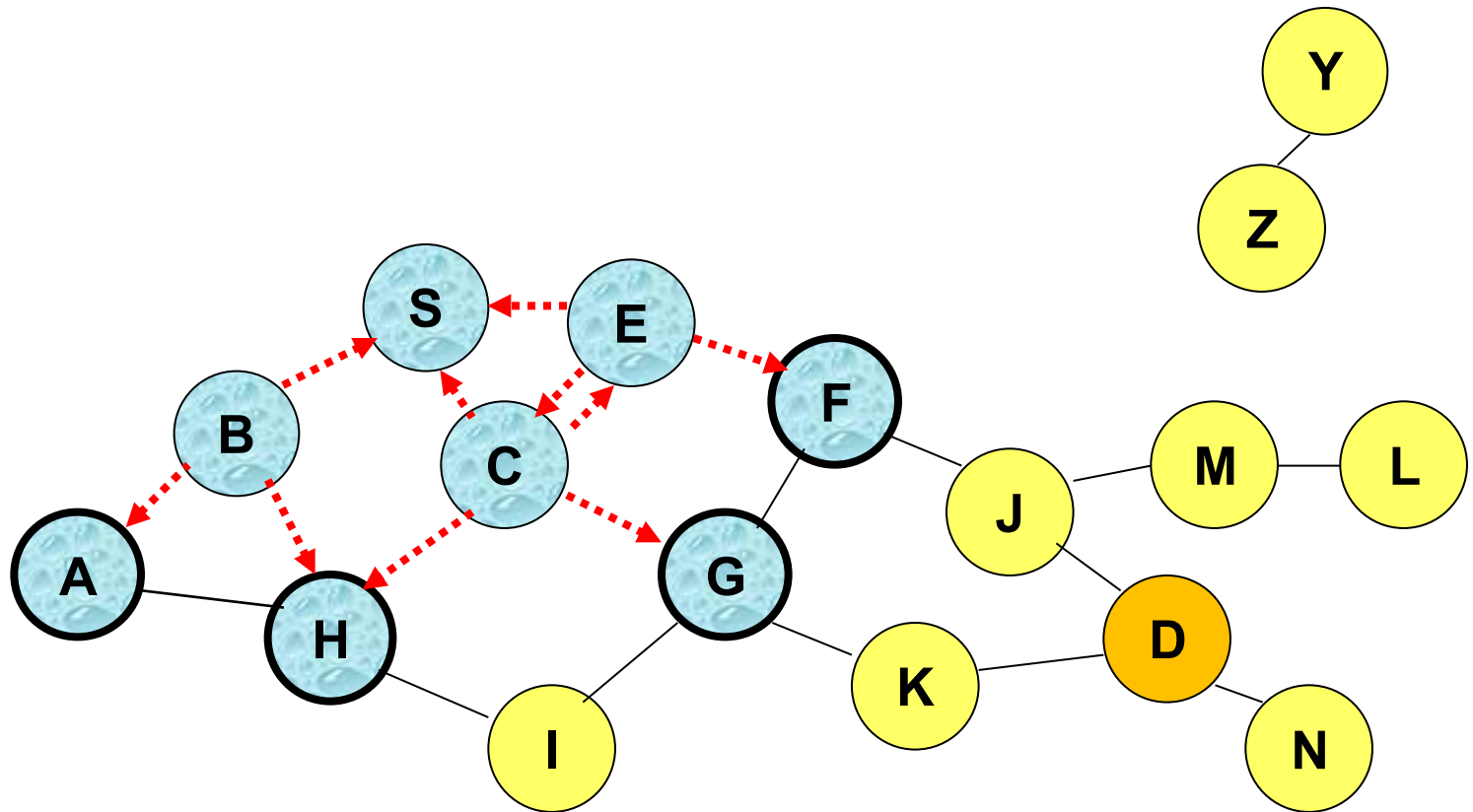


Represents a node that receives packet P for the first time



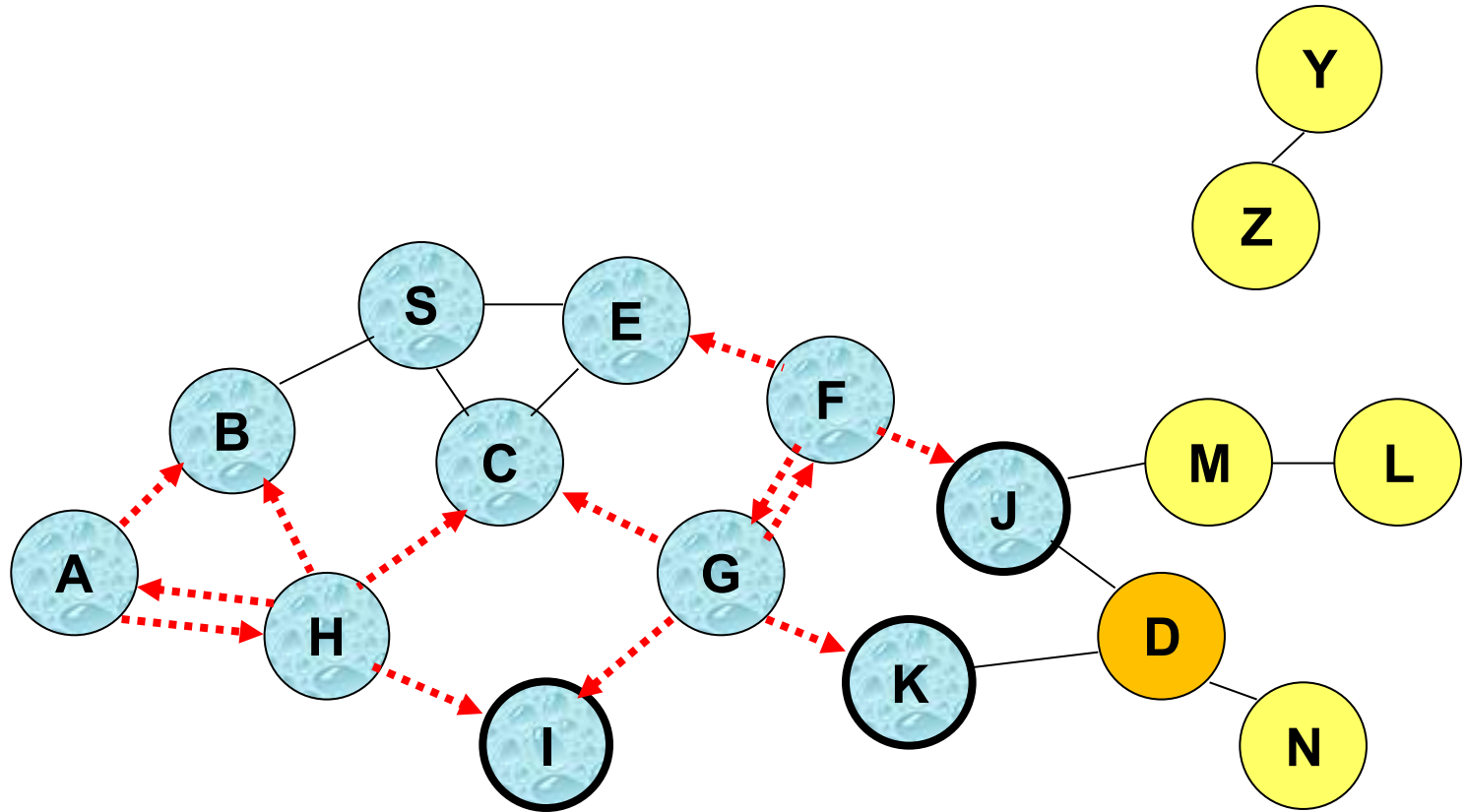
Represents transmission of packet P

Flooding for Data Delivery



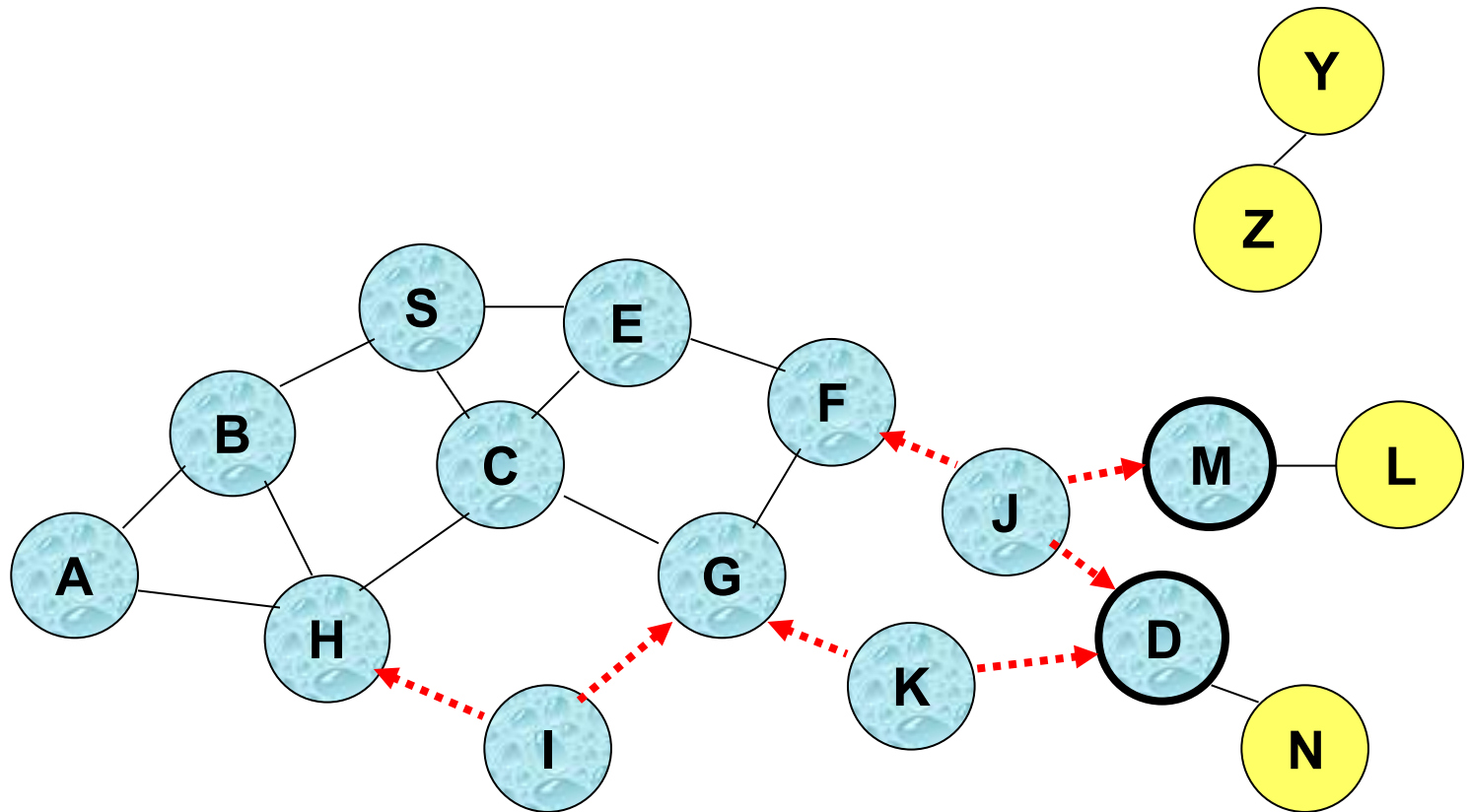
- Node H receives packet P from two neighbors:
potential for collision
- How to avoid collision?
 - CSMA/CA

Flooding for Data Delivery



- Node C receives packet P from G and H, but does not forward it again. Why?
 - because node C has already forwarded packet P once

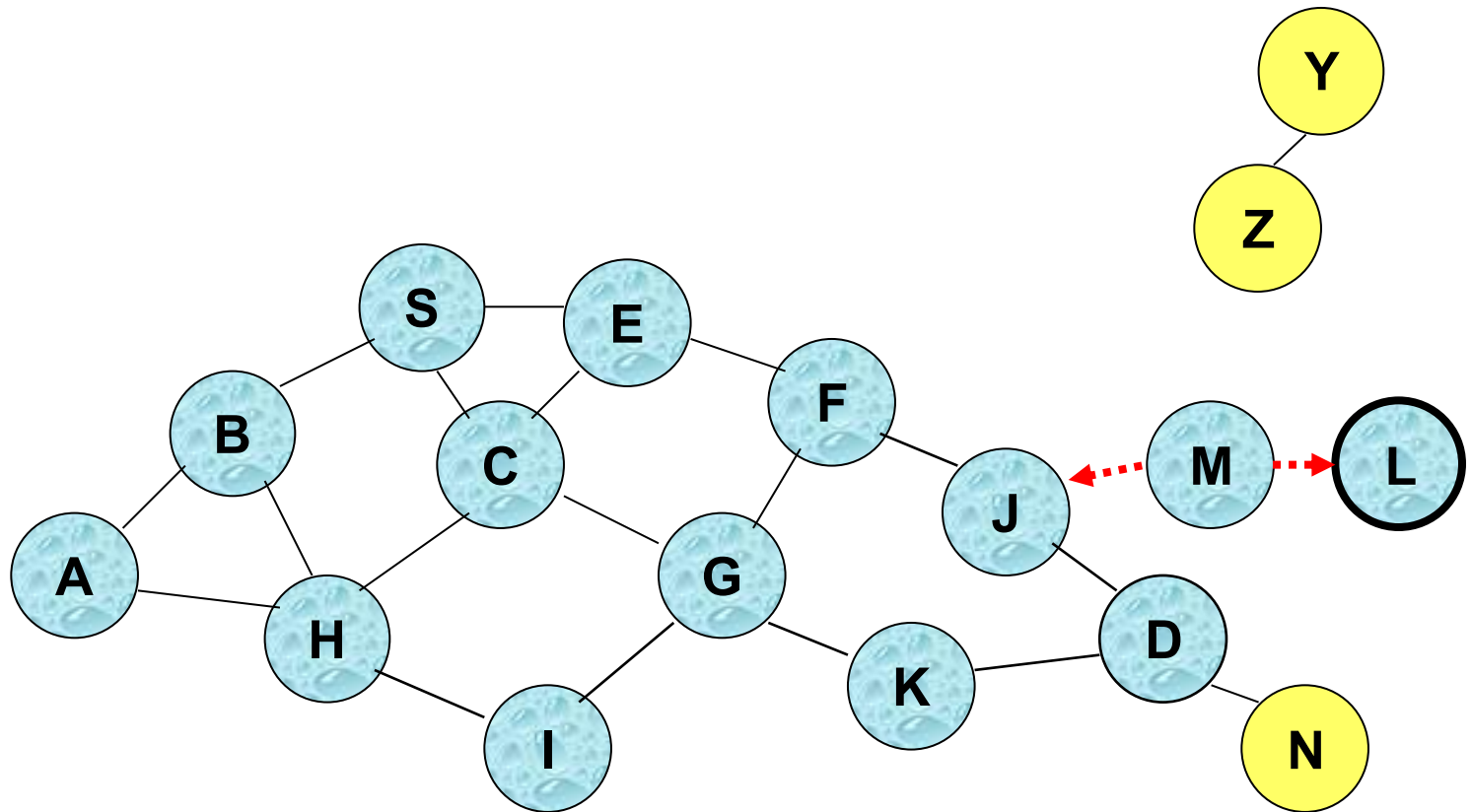
Flooding for Data Delivery



- Nodes J and K both broadcast packet P to node D
- Since nodes J and K are **hidden** from each other, their transmissions may collide

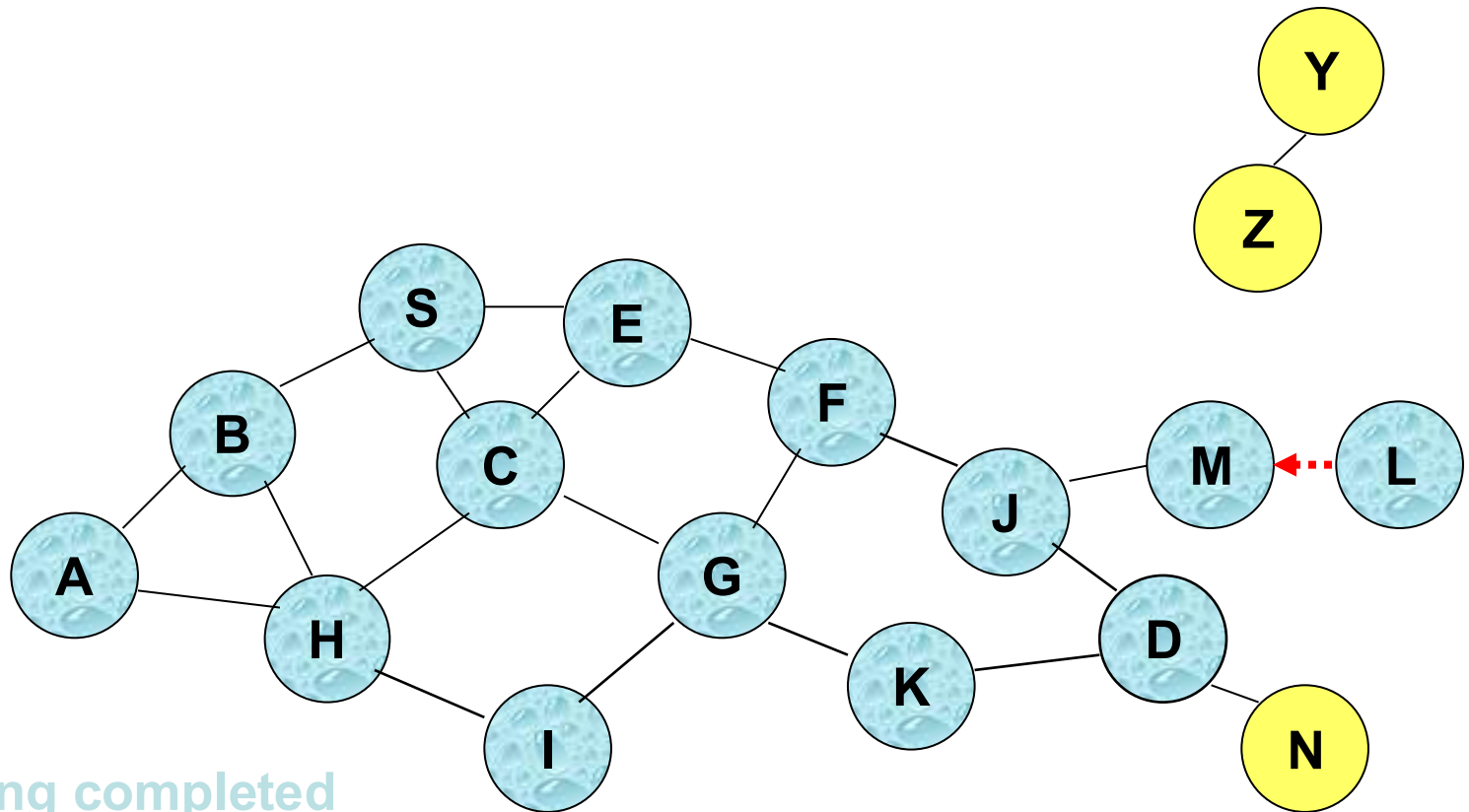
=> Packet P may not be delivered to node D at all, despite the use of flooding

Flooding for Data Delivery



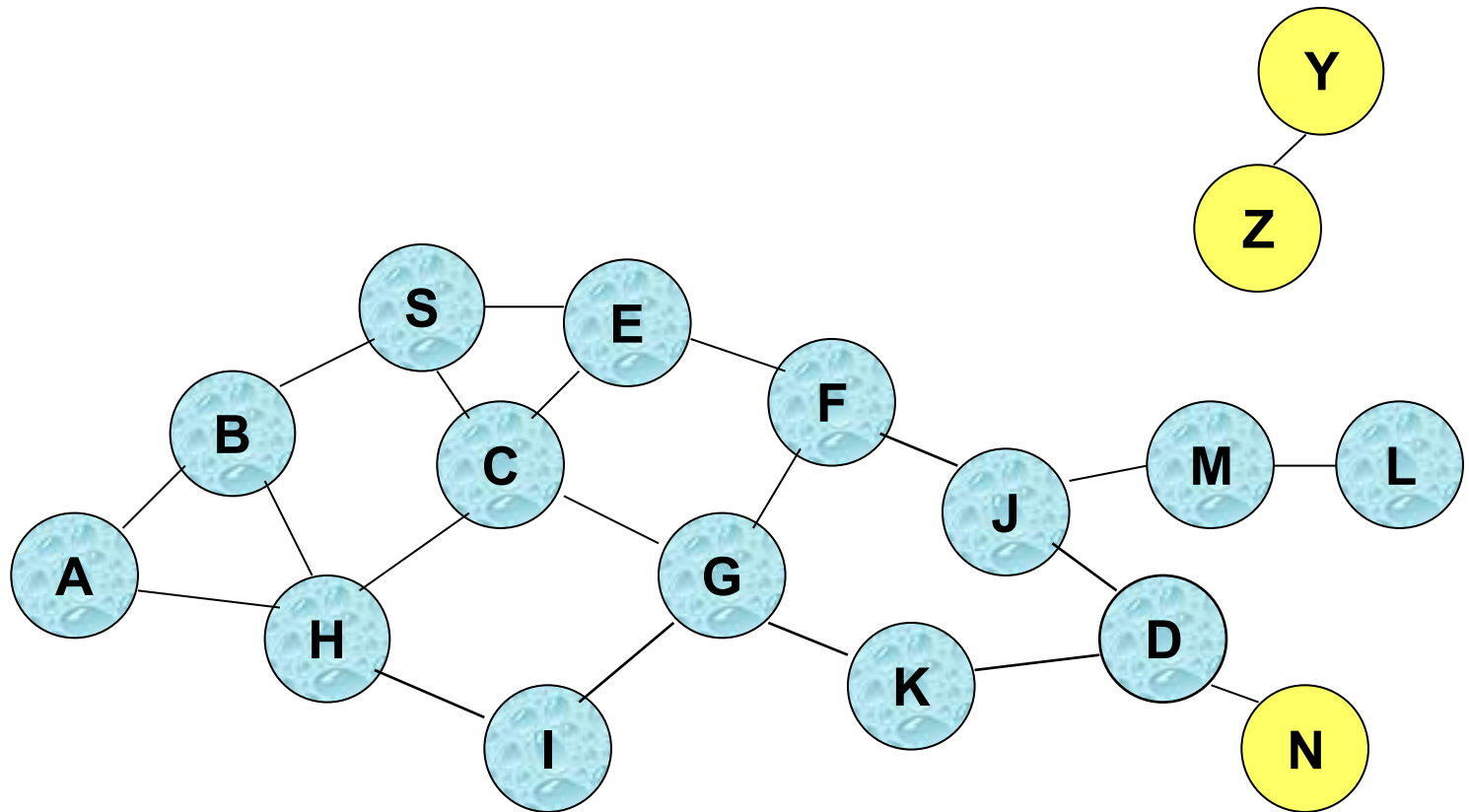
- Node D **does not forward** packet P, why?
because node D is the **intended destination** of packet P

Flooding for Data Delivery



- Flooding completed
- Nodes **unreachable** from S do not receive packet P (e.g., node Z)
- Nodes for which all paths from S go through the destination D₁₅ also do not receive packet P (example: node N)

Flooding for Data Delivery



- Flooding may deliver packets to too many nodes (in the **worst case**, all nodes reachable from sender may receive the packet)

Flooding for Data Delivery

- Sender S broadcasts data packet P to all its neighbors
- Each node receiving P forwards P to its neighbors
- **Sequence numbers field** used to avoid the possibility of forwarding the same packet more than once
- Packet P reaches destination D provided that D is reachable from sender S
- Node D does not forward the packet

Flooding for Data Delivery: Advantages

- Simplicity
- Efficient for:
 - Low information exchange rate
 - High mobility

In these cases, the overhead of explicit route discovery/maintenance incurred by other protocols may be relatively higher e.g. nodes transmit small data packets relatively infrequently, and topology changes occur between consecutive packet transmissions
- Potentially higher reliability of data delivery
 - Because packets may be delivered to the destination on multiple paths

Flooding for Data Delivery: Disadvantages

- Potentially, very high overhead
 - Data packets may be delivered to too many nodes who do not need to receive them
- Potentially lower reliability of data delivery (or higher delay)
 - Flooding uses broadcasting -- hard to implement reliable broadcast delivery without significantly increasing overhead
 - Broadcasting in IEEE 802.11 MAC is unreliable
 - In our example, nodes J and K may transmit to node D simultaneously, resulting in loss of the packet
 - in this case, destination would not receive the packet at all

Flooding of Control Packets

- Many protocols perform (potentially *limited*) flooding of control packets, instead of data packets
- The control packets are used to **discover/update routes** (and thus **establish/maintain** network state)
- Discovered routes are **subsequently used** to send data packet(s)
- Overhead of control packet flooding is **amortized** over data packets transmitted between consecutive control packet floods

Dynamic Source Routing (DSR)

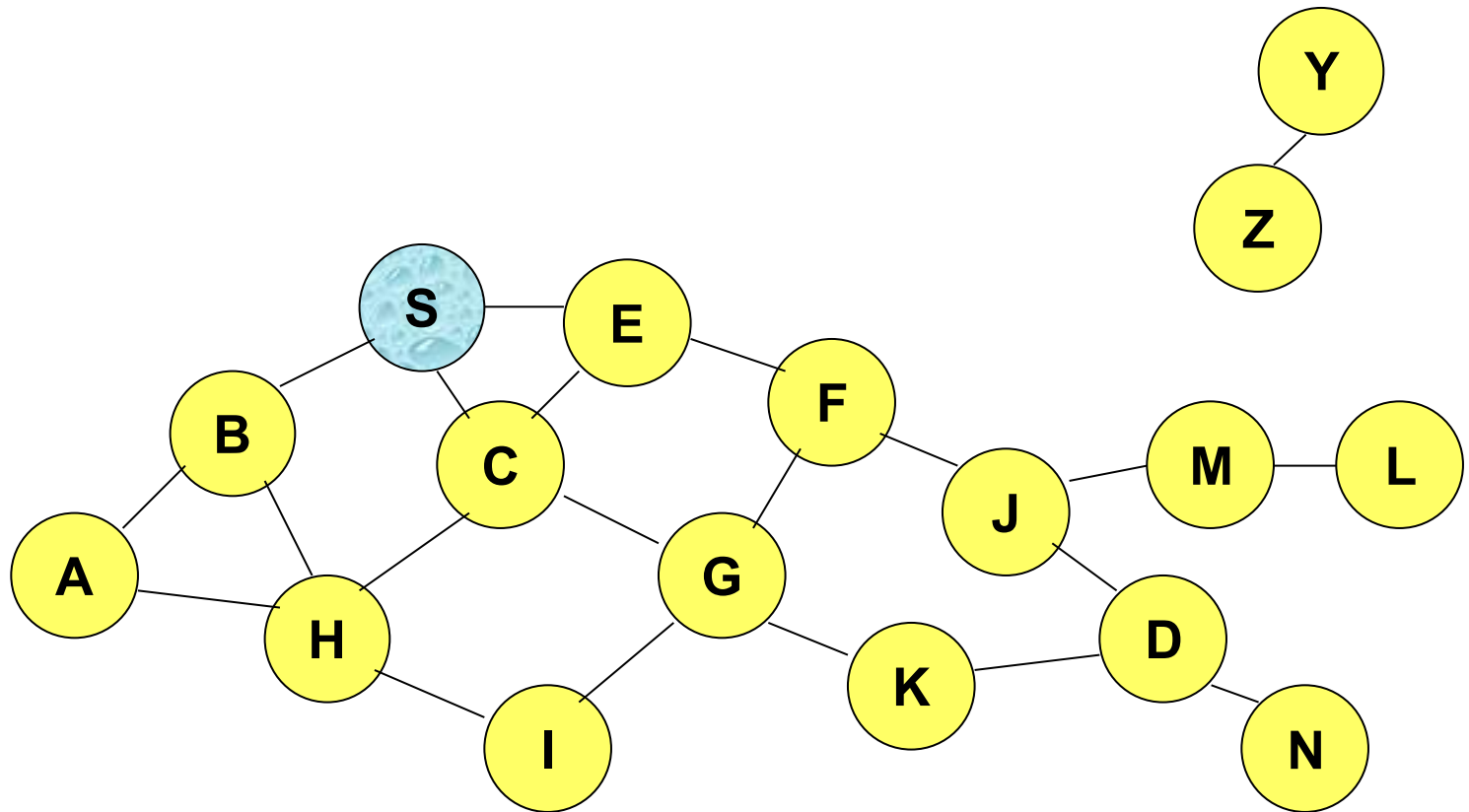
[Johnson96]

Dynamic Source Routing (DSR)

[Johnson96]

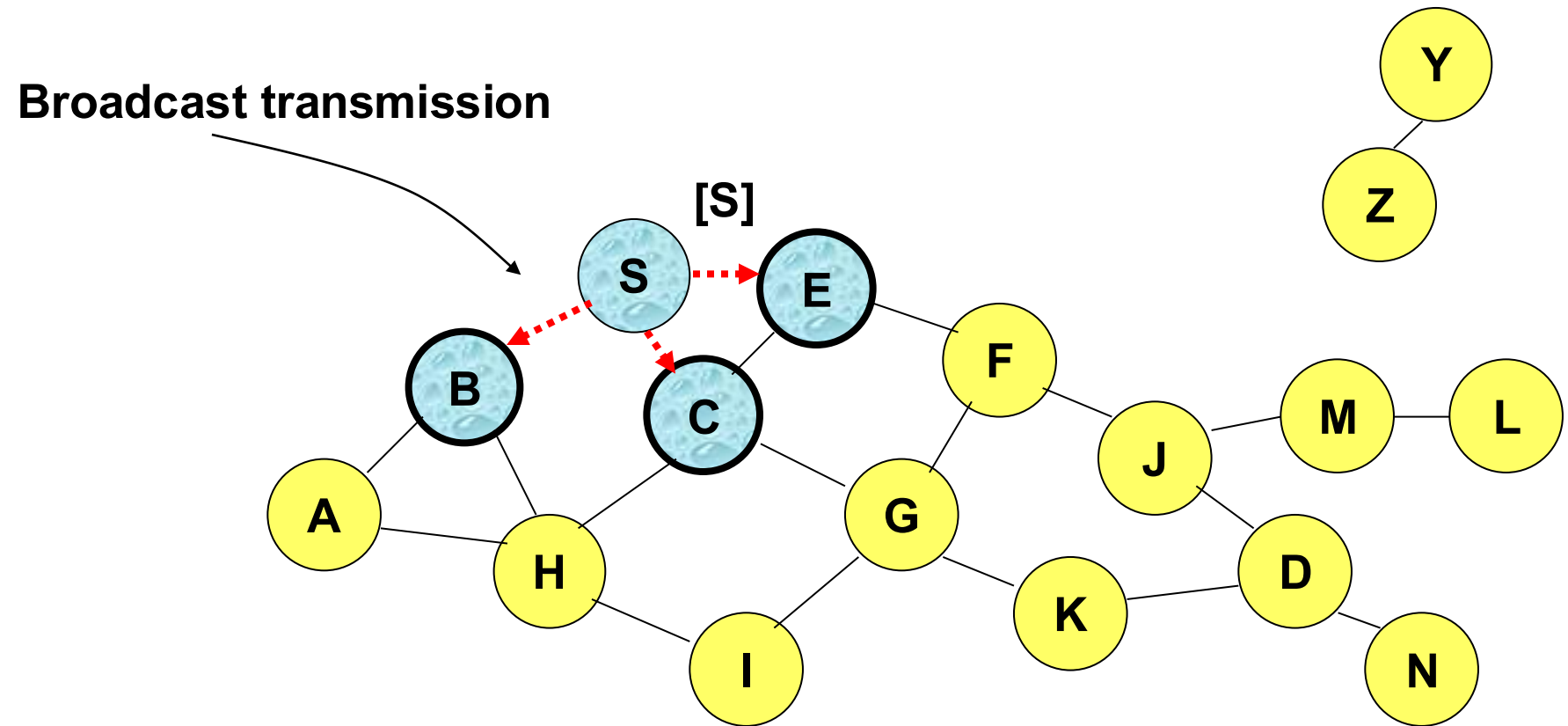
- When node S wants to send a packet to node D, but does not know a route to D, node S initiates a **route discovery**
- Source node S floods **Route Request (RREQ)**
- Each node **appends own identifier** when forwarding RREQ

Route Discovery in DSR



Represents a node that has received RREQ for D from S

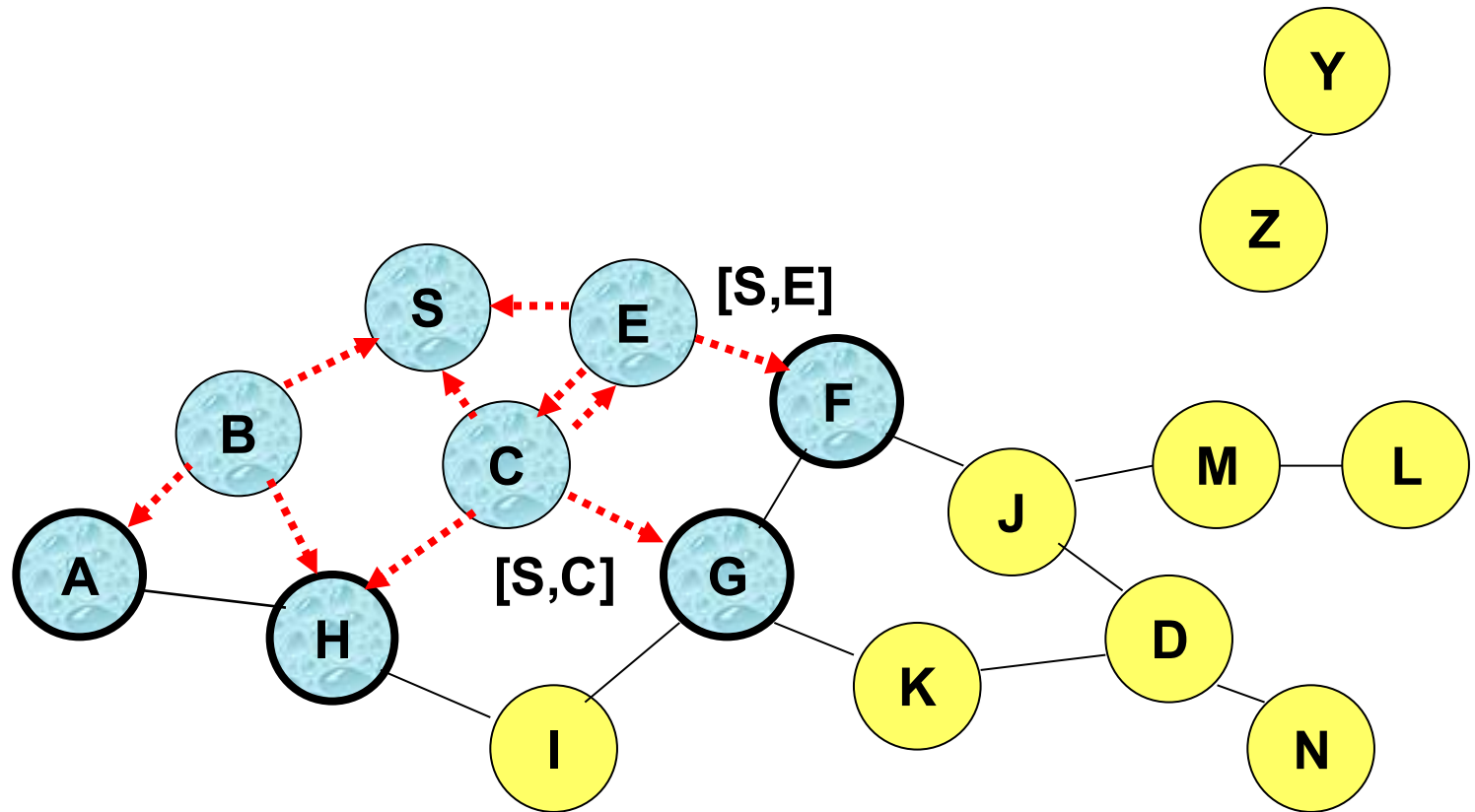
Route Discovery in DSR



.....→ Represents transmission of RREQ

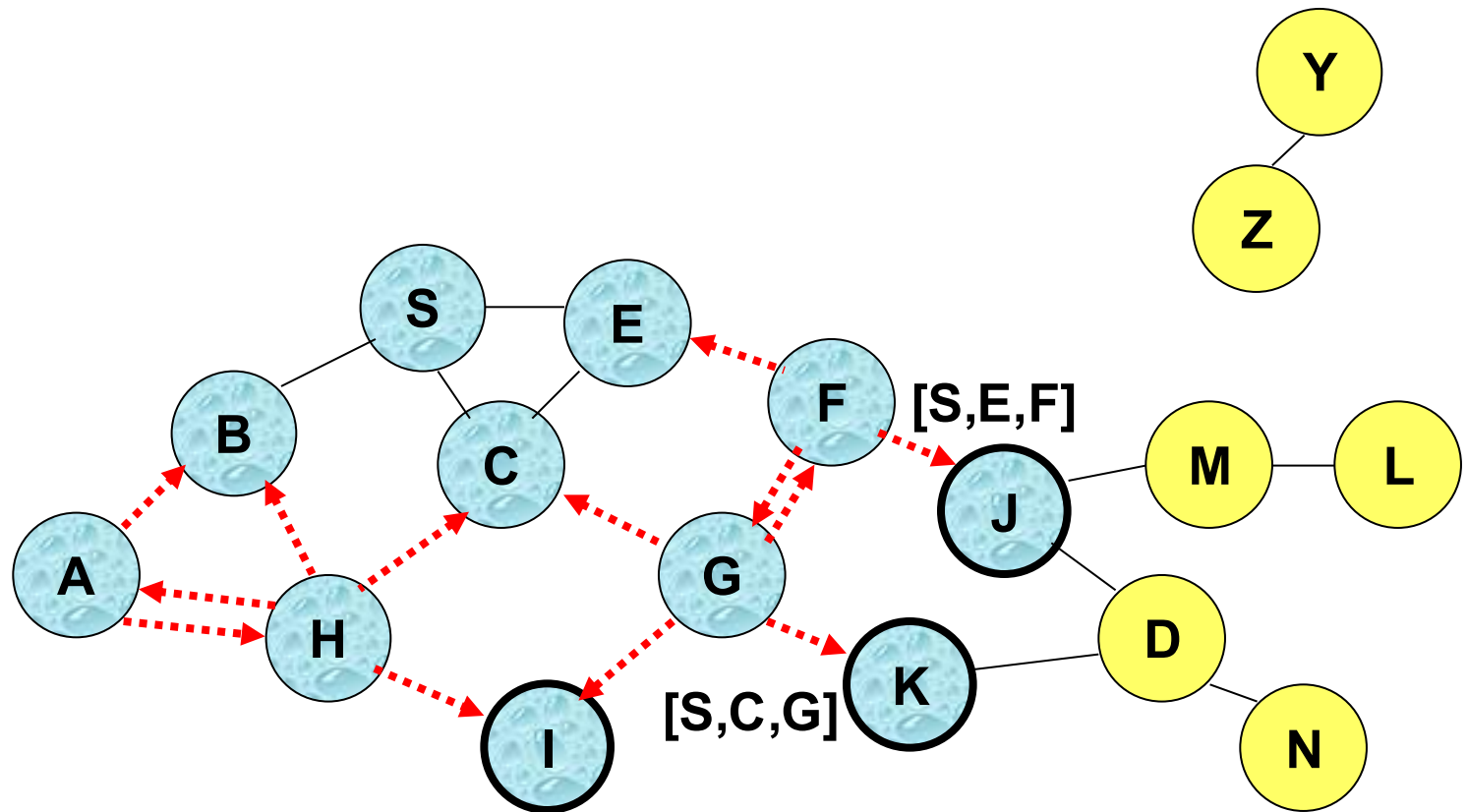
[X,Y] Represents list of identifiers appended to RREQ

Route Discovery in DSR



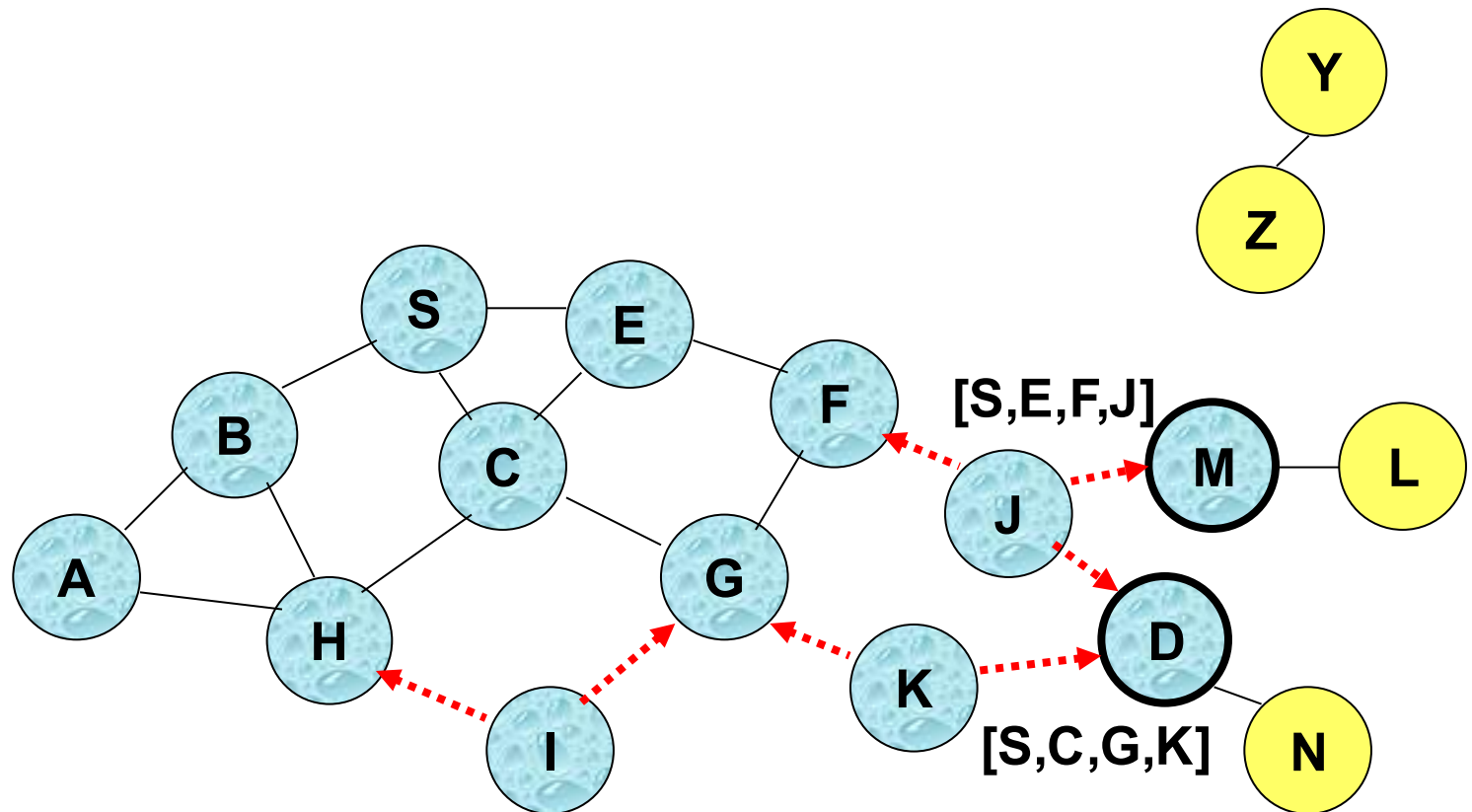
- Node H receives packet RREQ from two neighbors:
potential for collision

Route Discovery in DSR



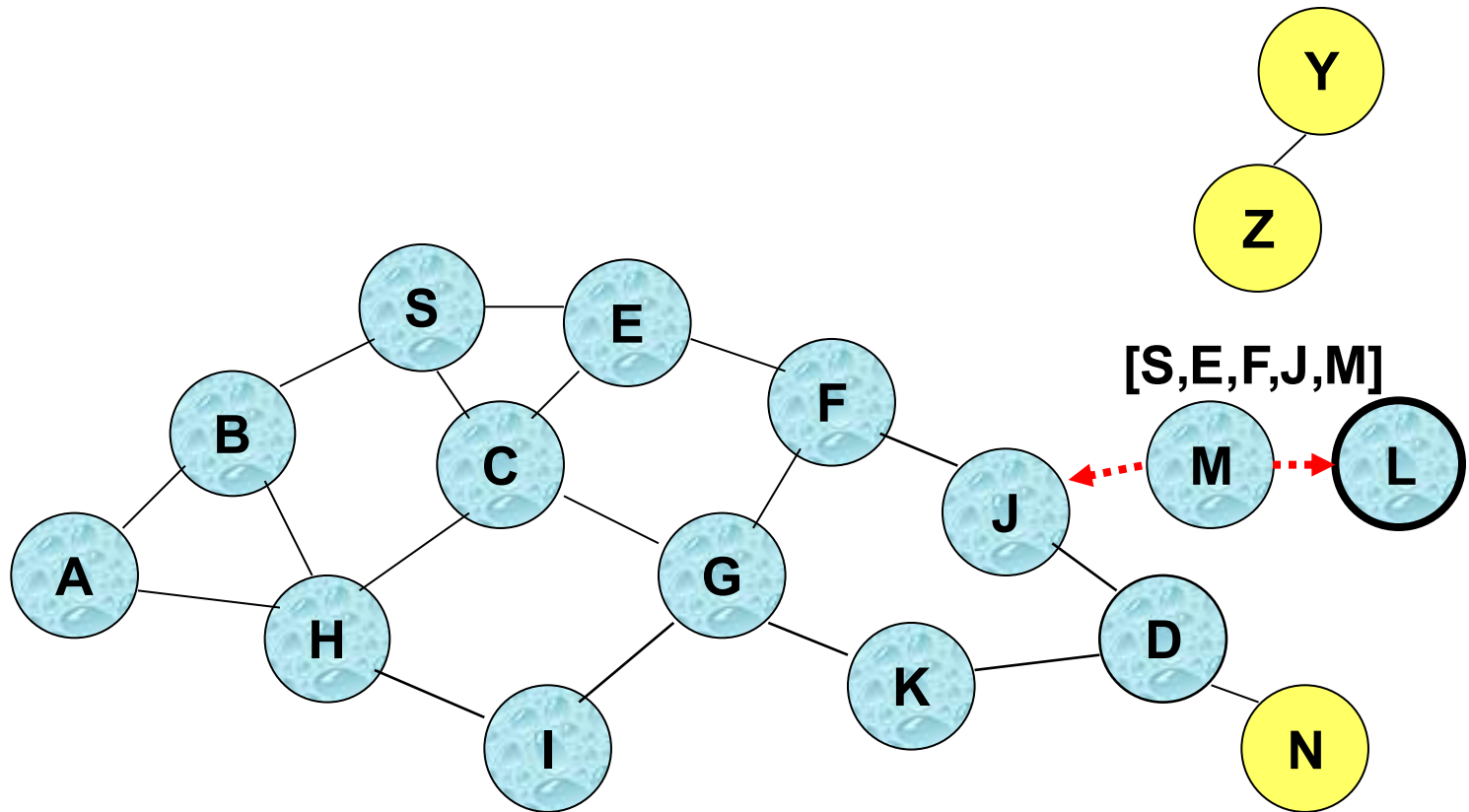
- Node C receives RREQ from G and H, but does not forward it again, because node C has **already forwarded RREQ** once

Route Discovery in DSR



- Nodes J and K both broadcast RREQ to node D
- Since nodes J and K are **hidden** from each other, their **transmissions may collide**

Route Discovery in DSR

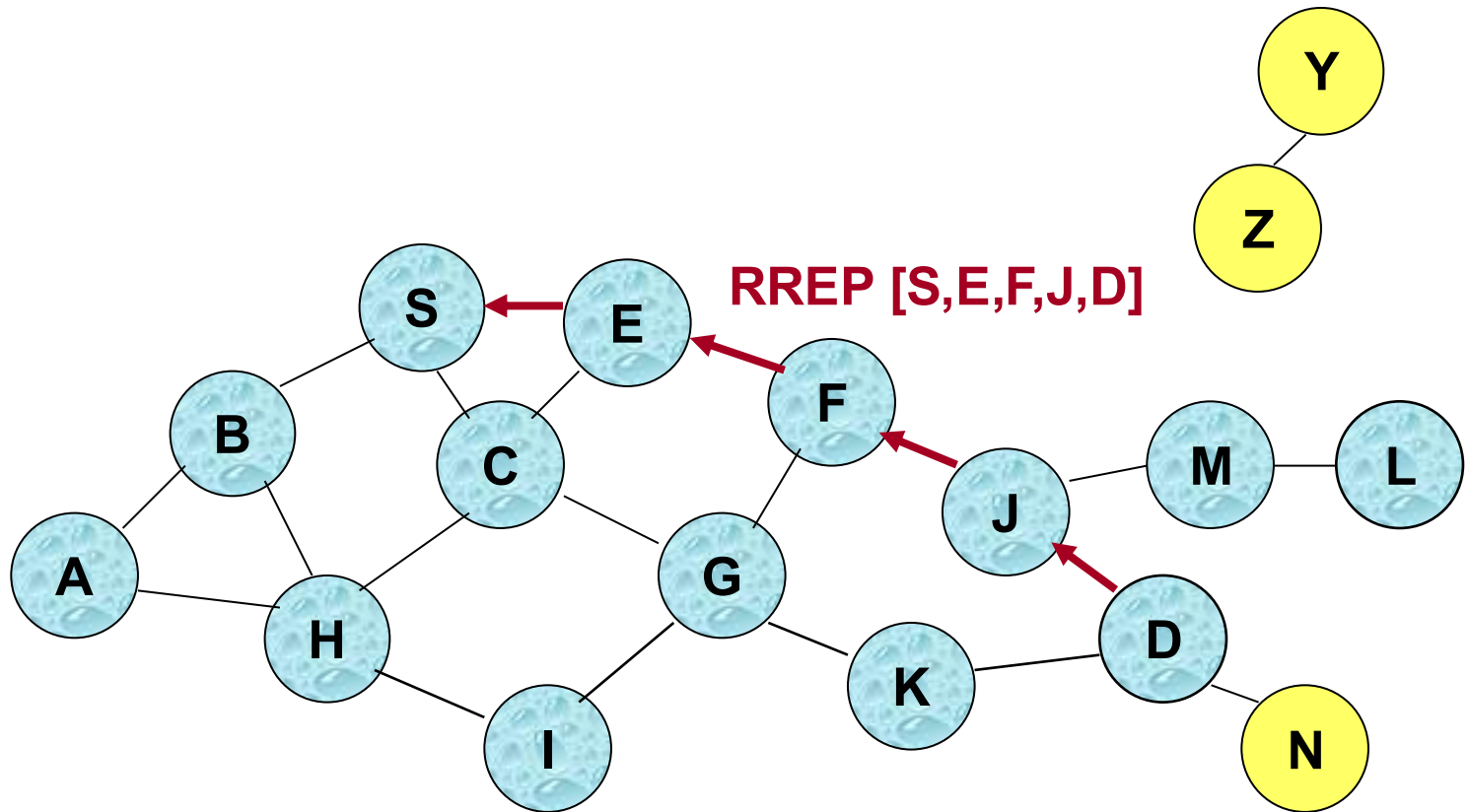


- Node D **does not forward** RREQ, because node D is the **intended target** of the route discovery

Route Discovery in DSR

- How does Source S know about the route?
- Destination D on receiving the first RREQ, sends a **Route Reply (RREP)**
- RREP is sent on a route obtained by **reversing** the route appended to received RREQ
- RREP packet **contains the route** from S to D that was discovered using the RREQ packet

Route Reply in DSR

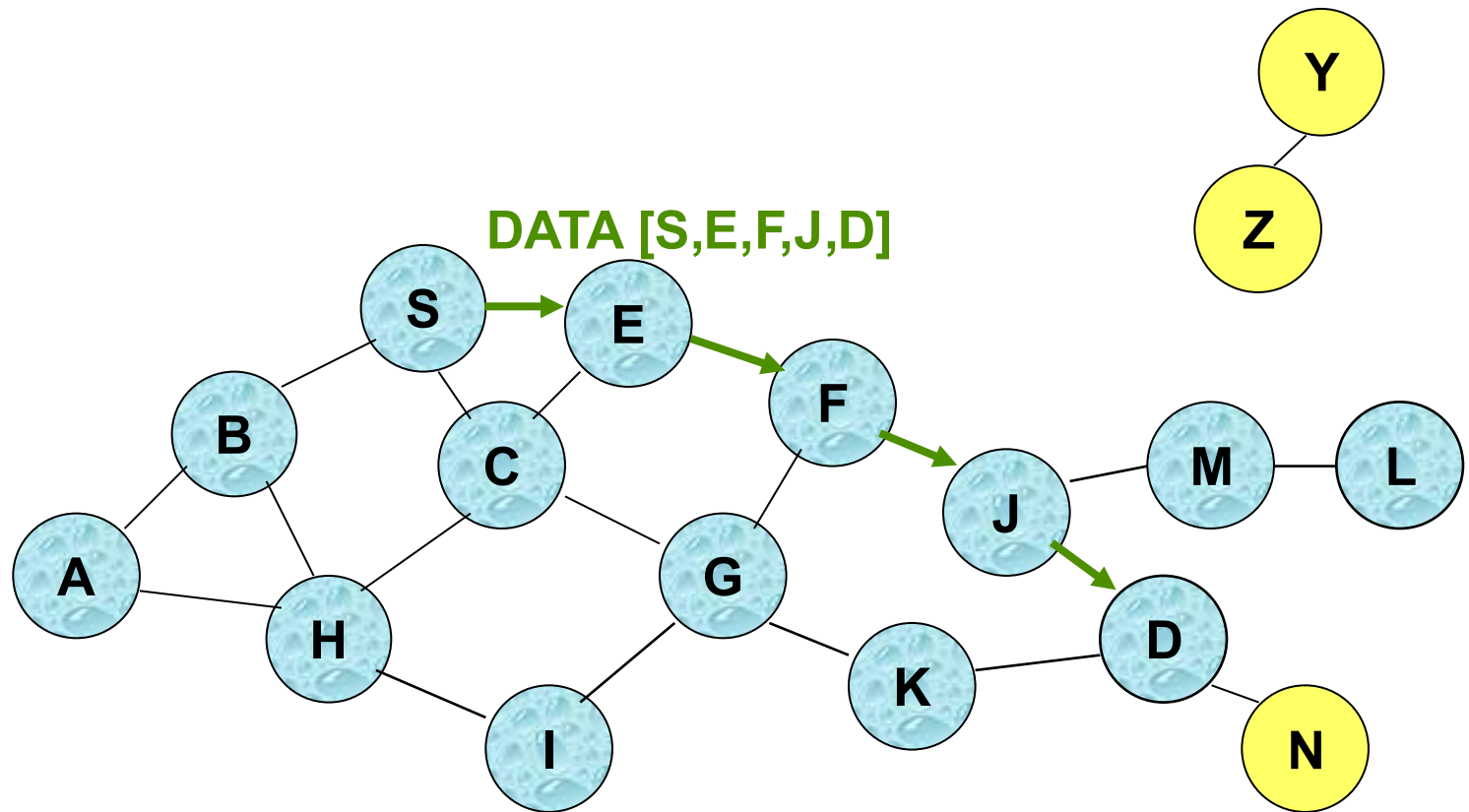


← Represents RREP control message

Dynamic Source Routing (DSR)

- Node S on receiving RREP, **caches** the route included in the RREP
- When node S subsequently sends a data packet to D, the entire route is included in the packet header
 - **hence the name source routing**
- Intermediate nodes use the **source route** included in a packet to determine to whom a packet should be forwarded

Data Delivery in DSR



Packet header size grows with route length

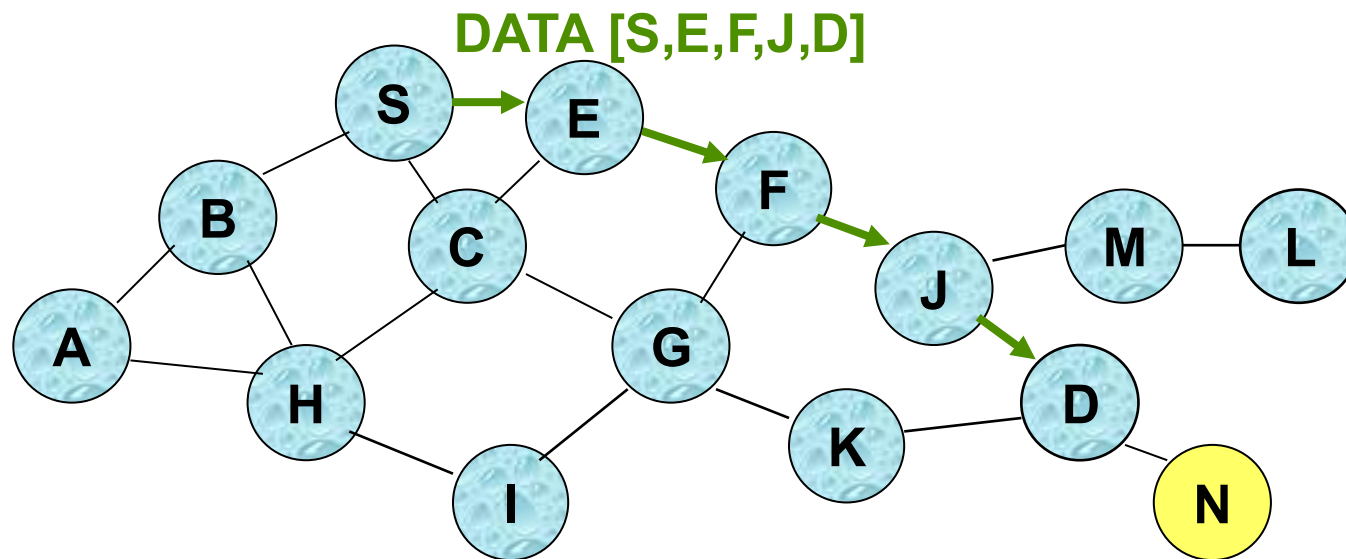
When to Perform a Route Discovery?

- When node S wants to send data to node D (i.e. *on-demand*), but does not know a valid route to node D

How to reduce the overhead?

DSR Optimization: Route Caching

- Each node caches a new route it learns by *any means*
- When node S finds route [S,E,F,J,D] to node D, node S also learns route [S,E,F] to node F



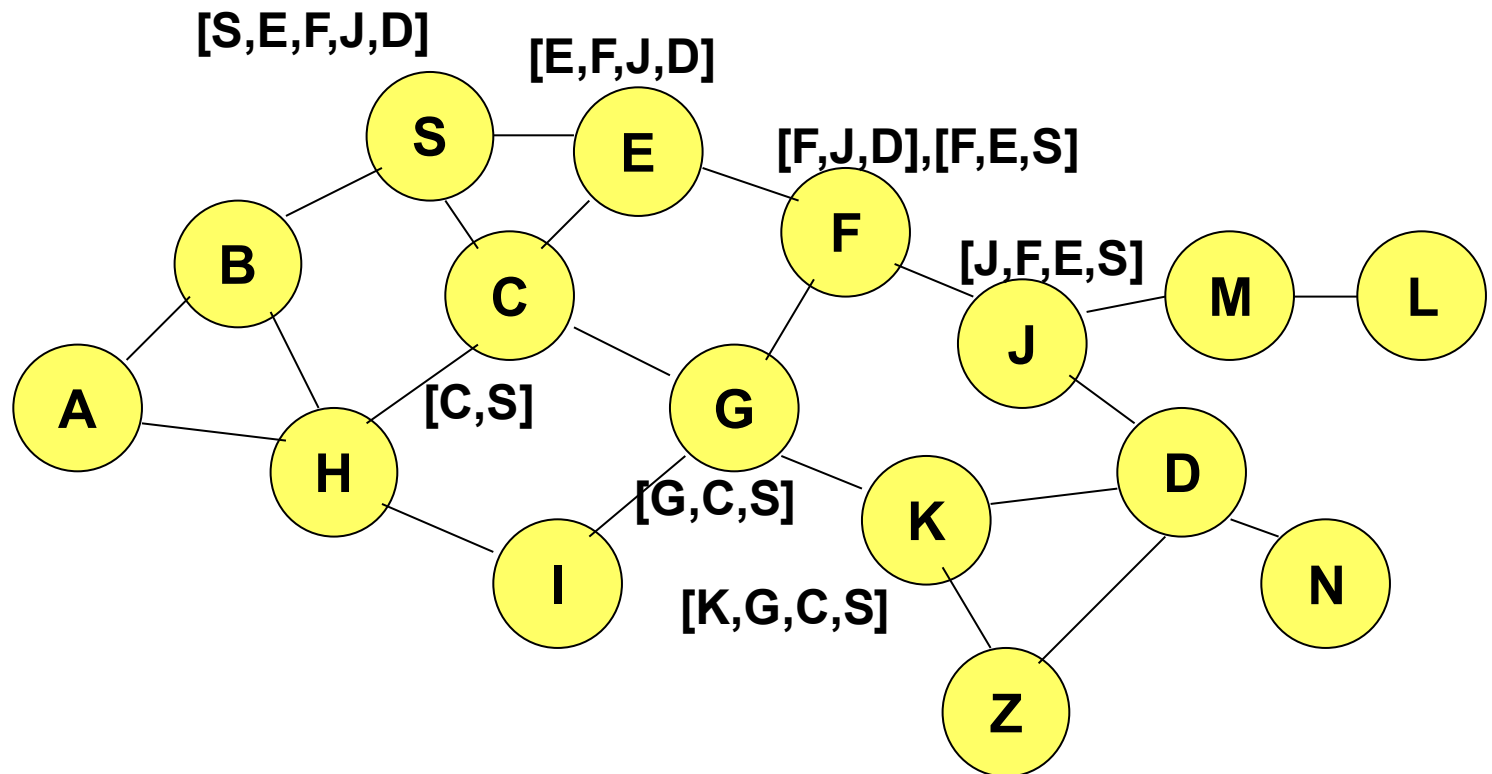
DSR Optimization: Route Caching

- Each node caches a new route it learns by *any means*
- When node S finds route [S,E,F,J,D] to node D, node S also learns route [S,E,F] to node F
- When node K receives Route Request [S,C,G] destined for node D, node K learns route [K,G,C,S] to node S
- When node F forwards Route Reply [S,E,F,J,D], node F learns route [F,J,D] to node D
- When node E forwards Data [S,E,F,J,D] it learns route [E,F,J,D] to node D
- A node may also learn a route when it overhears Data packets, even if it is not forwarding!

Use of Route Caching

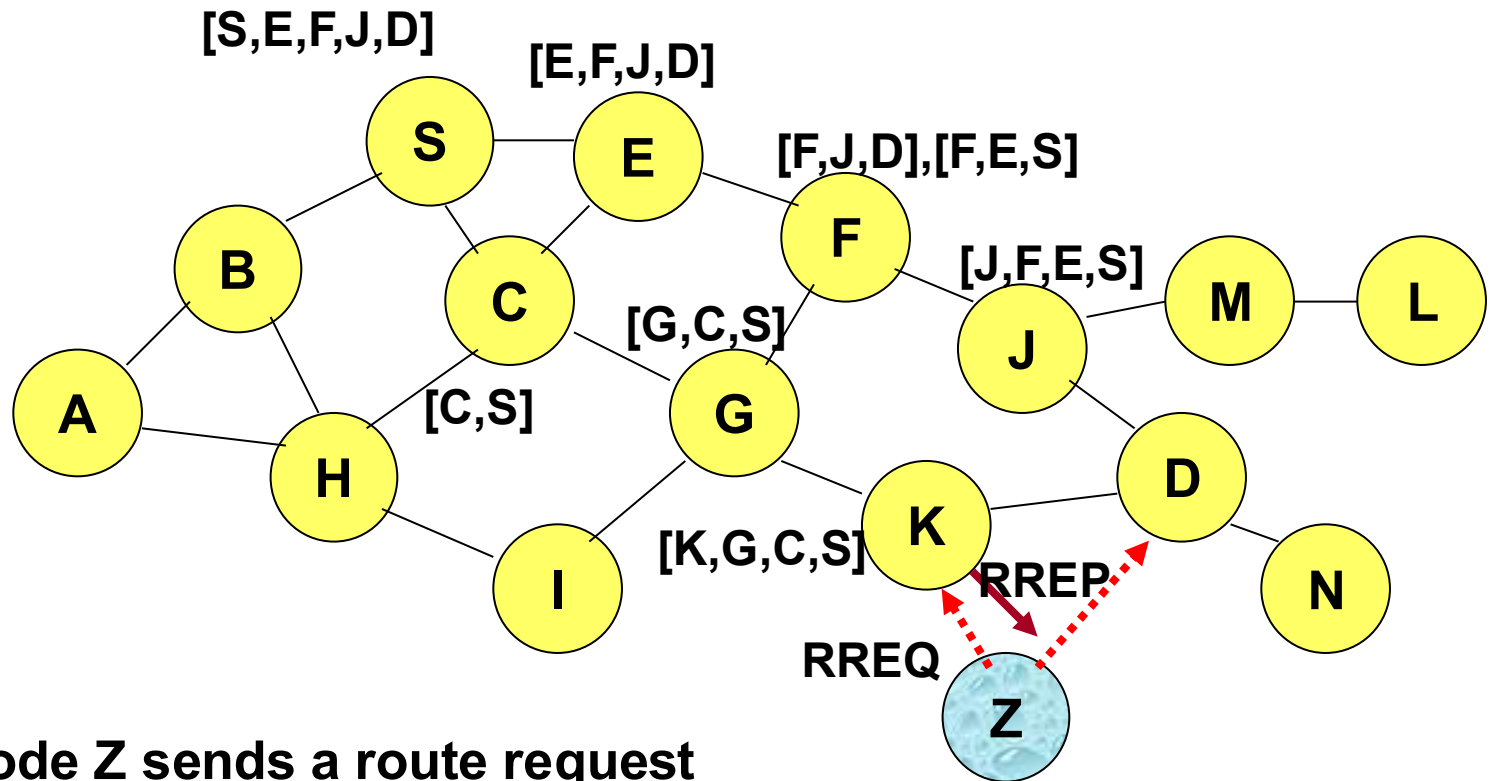
- When node S learns that a route to node D is broken, it uses another route from its local cache, if such a route to D exists in its cache. Otherwise, node S initiates route discovery by sending a route request
- Node X on receiving a Route Request for some node D can send a Route Reply if node X knows a route to node D
- Use of route cache
 - can speed up route discovery
 - can reduce propagation of route requests

Use of Route Caching



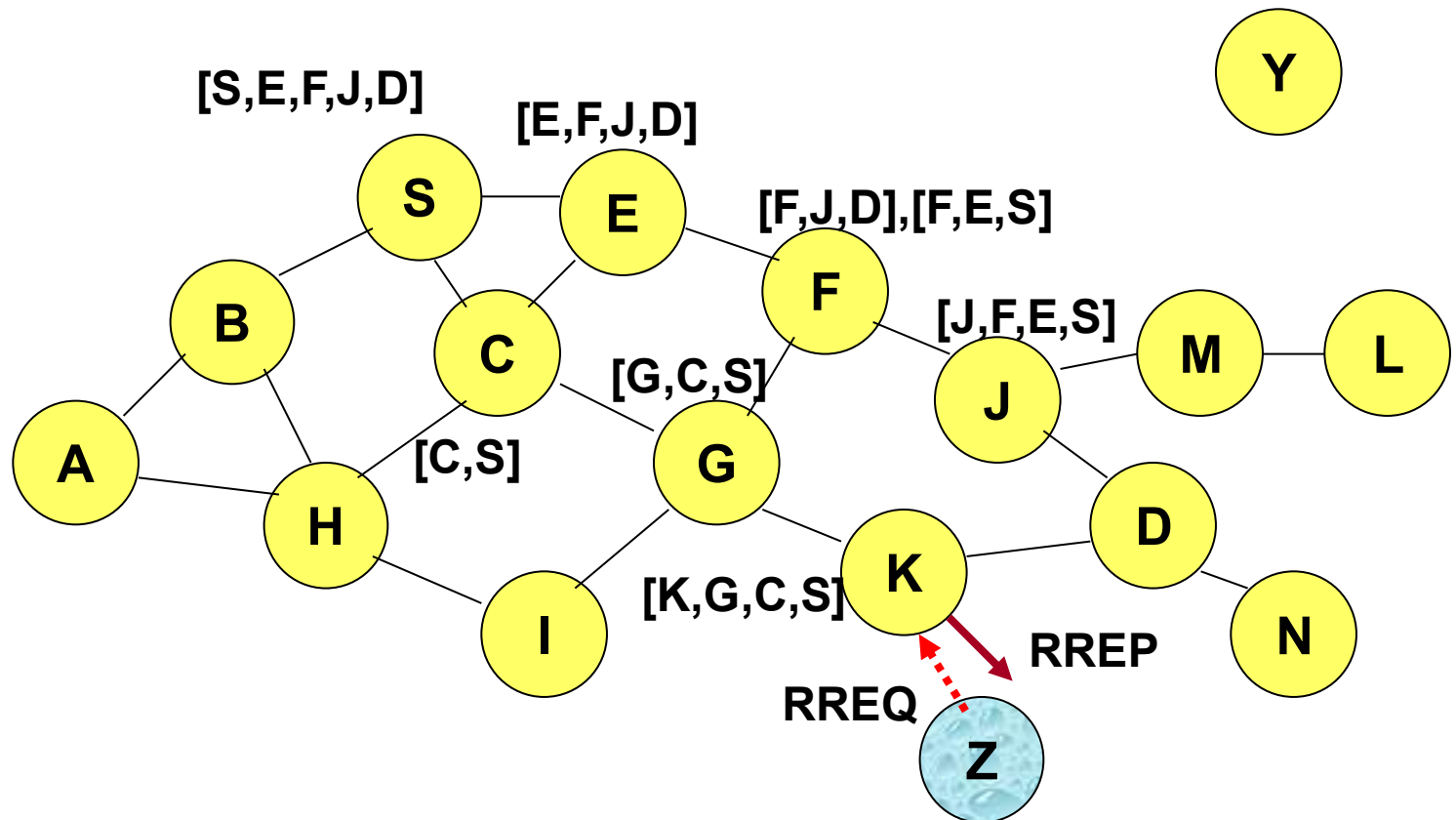
[P,Q,R] Represents cached route at a node

Use of Route Caching: Can Speed up Route Discovery



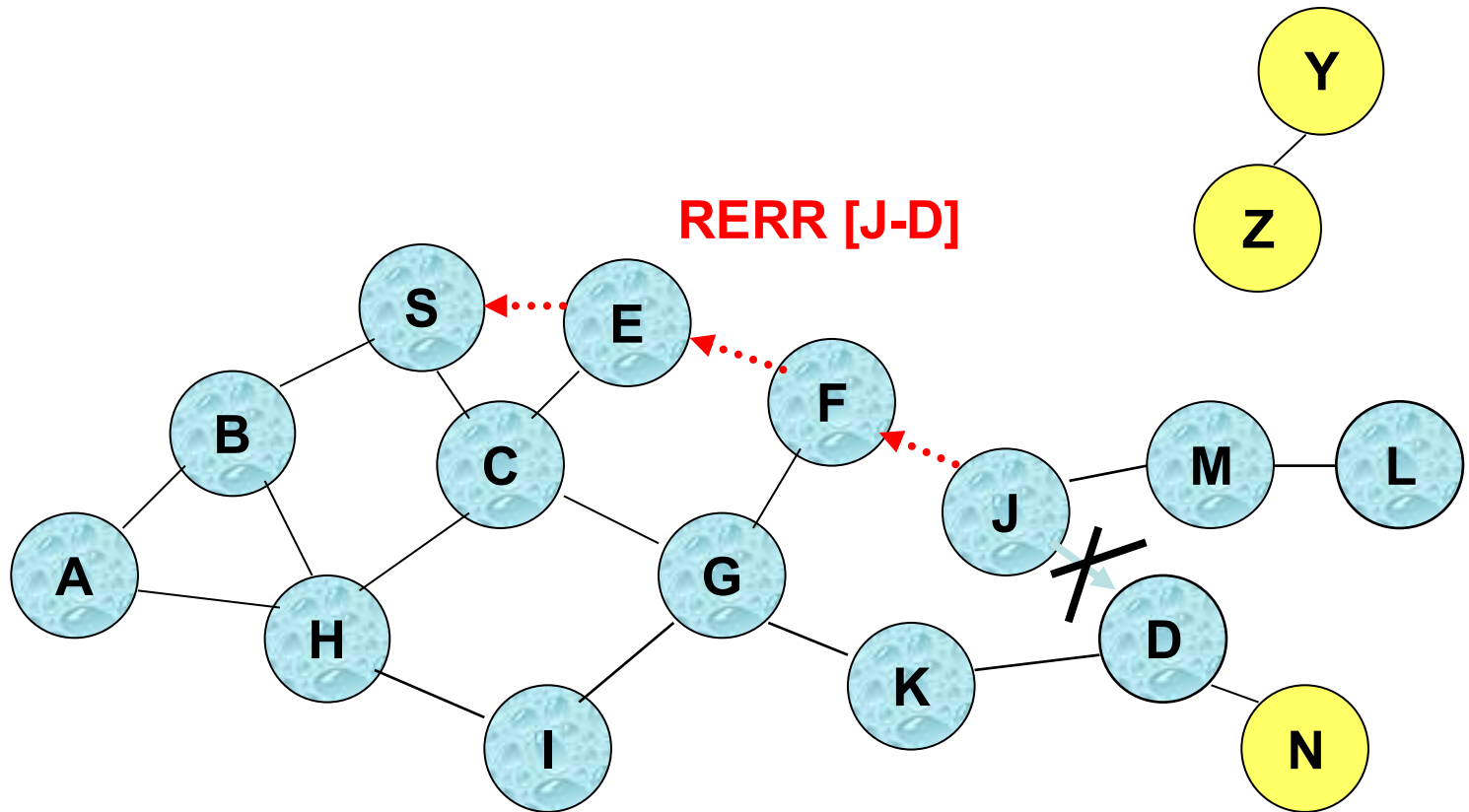
When node Z sends a route request for node C, node K sends back a route reply [Z,K,G,C] to node Z using a locally cached route

Use of Route Caching: Can Reduce Propagation of Route Requests



Assume that there is no link between D and Z.
Route Reply (RREP) from node K **limits flooding** of RREQ.
In general, the reduction may be less dramatic.

Route Error (RERR)



J sends a route error to S along route J-F-E-S when its attempt to forward the data packet S (with route SEFJD) on J-D fails

Nodes hearing RERR update their route cache to remove link J-D⁴⁰

Route Caching: Disadvantages

- Stale caches can adversely affect performance
- With passage of time and host mobility, cached routes may become invalid
- A sender host may try several stale routes (obtained from local cache, or replied from cache by other nodes), before finding a good route
- (An illustration of the adverse impacts e.g., on TCP, can be found in [Holland99])

Dynamic Source Routing: Advantages

- Routes maintained only between nodes who need to communicate (ie. *on-demand*)
 - reduces overhead of route maintenance
- Route caching can further reduce route discovery overhead
- A single route discovery may yield many routes to the destination, due to multiple intermediate nodes replying from local caches

Dynamic Source Routing: Disadvantages

- Packet header size grows with route length due to source routing
- Flood of route requests may potentially reach all nodes in the network
- Care must be taken to avoid collisions between route requests propagated by neighboring nodes
 - insertion of random delays before forwarding RREQ

Dynamic Source Routing: Disadvantages (cache)

- Increased contention if too many route replies come back due to nodes replying using their local cache
 - **Route Reply Storm problem**
 - Reply storm may be eased by preventing a node from sending RREP if it hears another RREP with a shorter route
- An intermediate node may send Route Reply using a stale cached route, thus polluting other caches
- This problem can be eased if some mechanism to purge (potentially) invalid cached routes is incorporated.
- For some proposals for cache invalidation, see **[Hu00Mobicom]**
 - Static timeouts
 - Adaptive timeouts based on link stability

Ad Hoc On-Demand Distance Vector Routing (AODV)

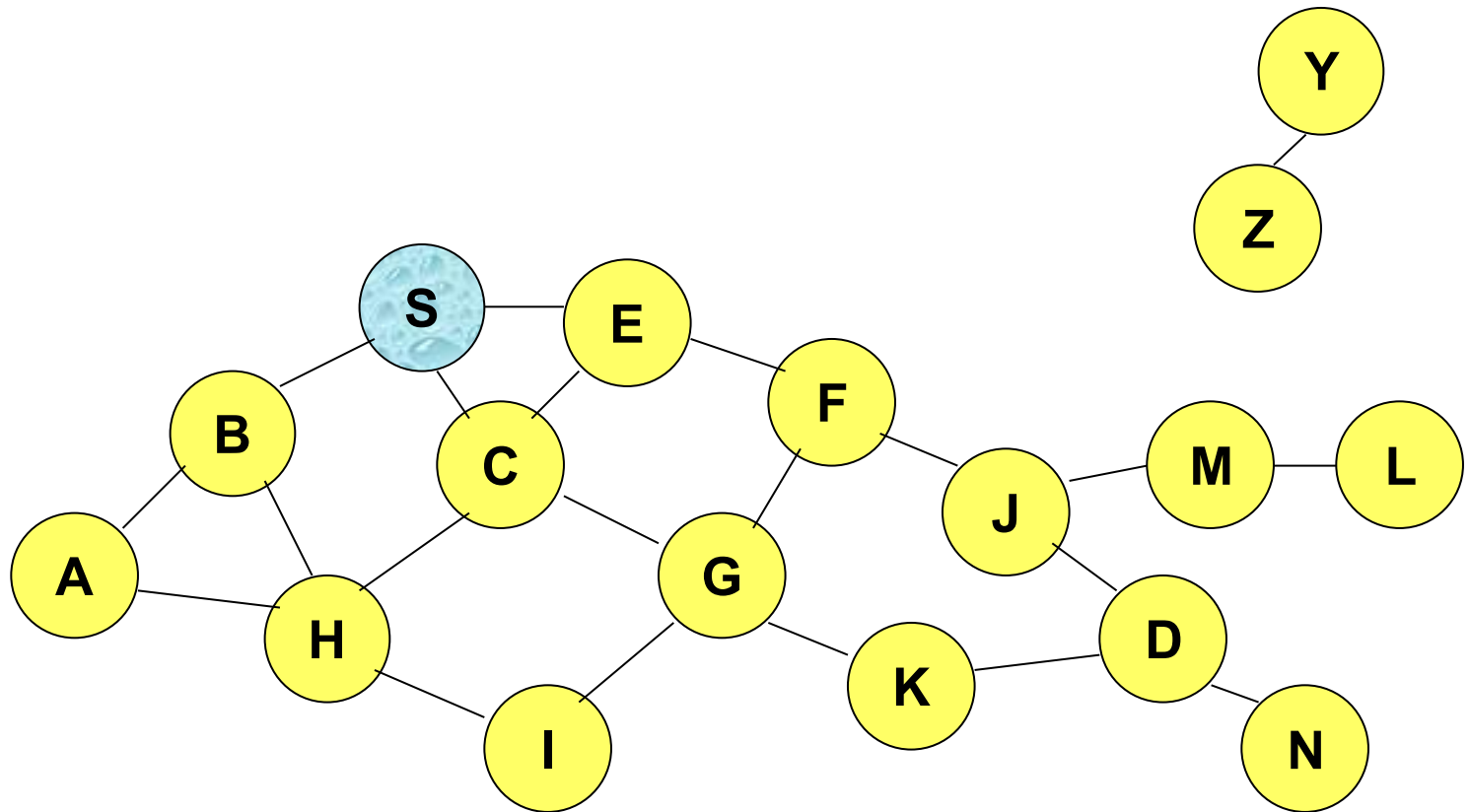
Ad Hoc On-Demand Distance Vector Routing (AODV)

- DSR includes source routes in packet headers
- Resulting large headers can sometimes degrade performance
 - particularly when data contents of a packet are small
- AODV attempts to improve on DSR by maintaining routing tables at the nodes, so that data packets do not have to contain routes
- AODV retains the desirable feature of DSR that routes are maintained only between nodes which need to communicate (on-demand)

AODV

- Route Requests (**RREQ**) are forwarded in a manner similar to DSR
- When a node re-broadcasts a Route Request, it **sets up a reverse path** pointing towards the source
 - AODV assumes symmetric (bi-directional) links
- When the intended destination receives a Route Request, it replies by sending a Route Reply (**RREP**)
- Route Reply **travels along the reverse path** set-up when Route Request is forwarded

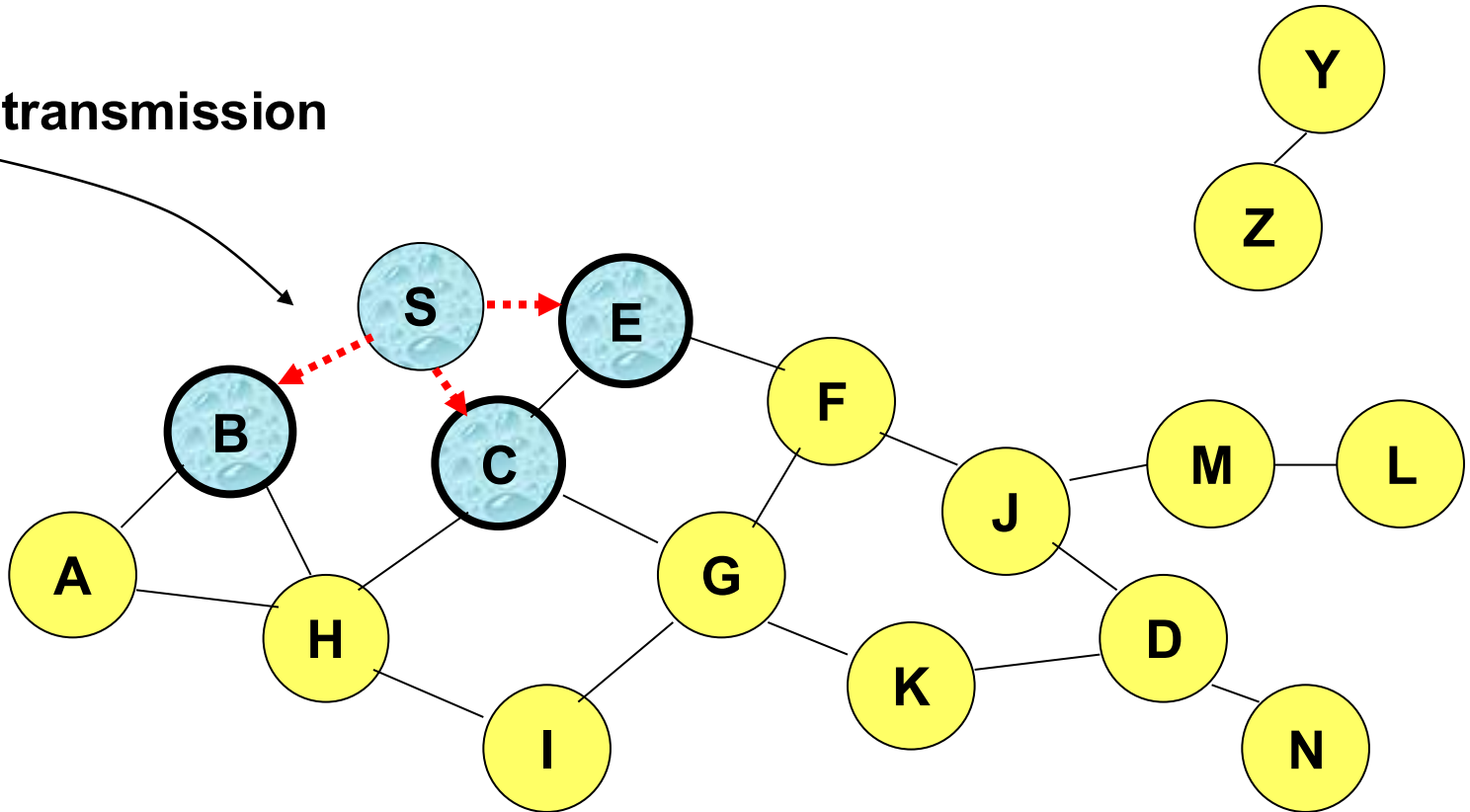
Route Requests in AODV



Represents a node that has received RREQ for D from S

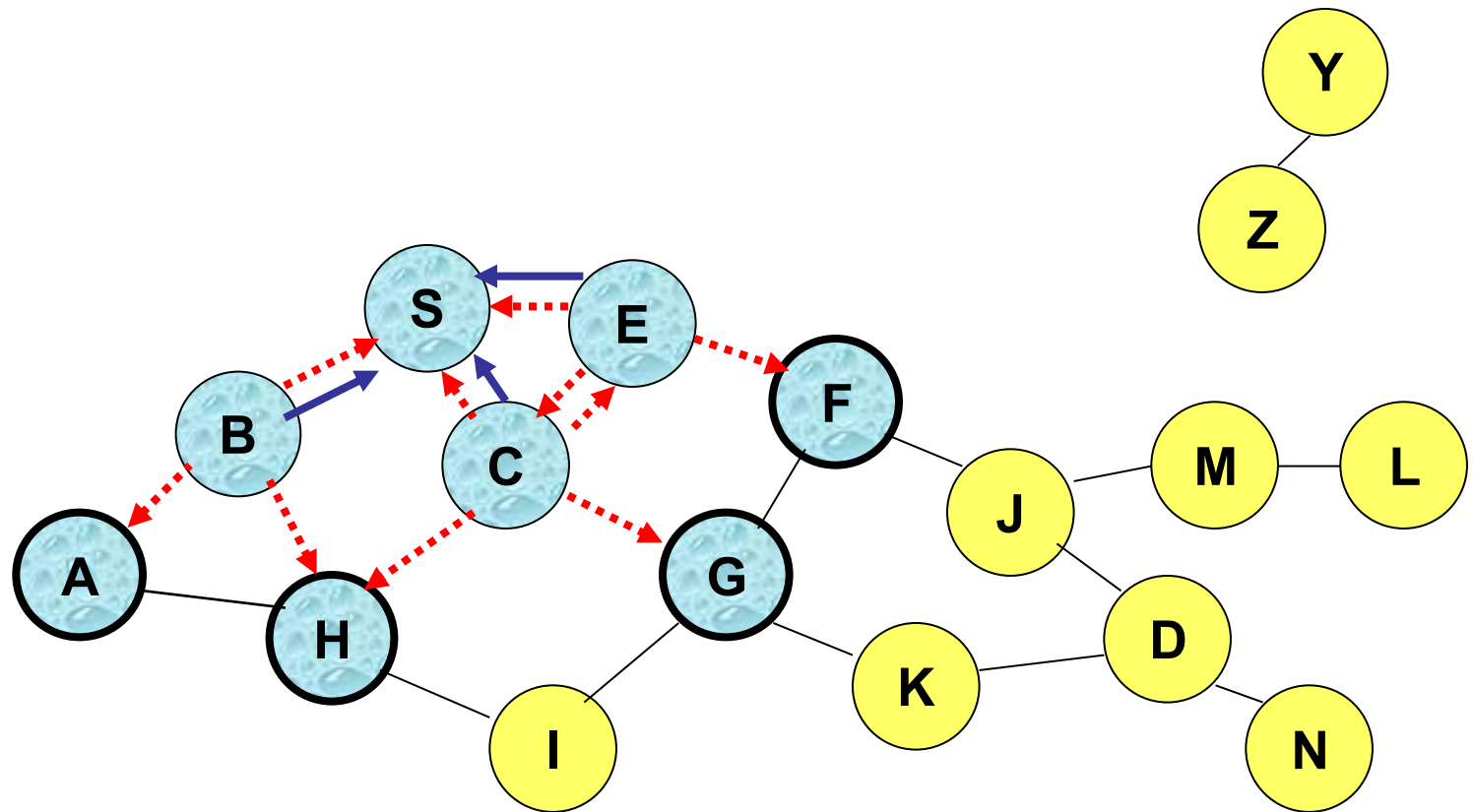
Route Requests in AODV

Broadcast transmission



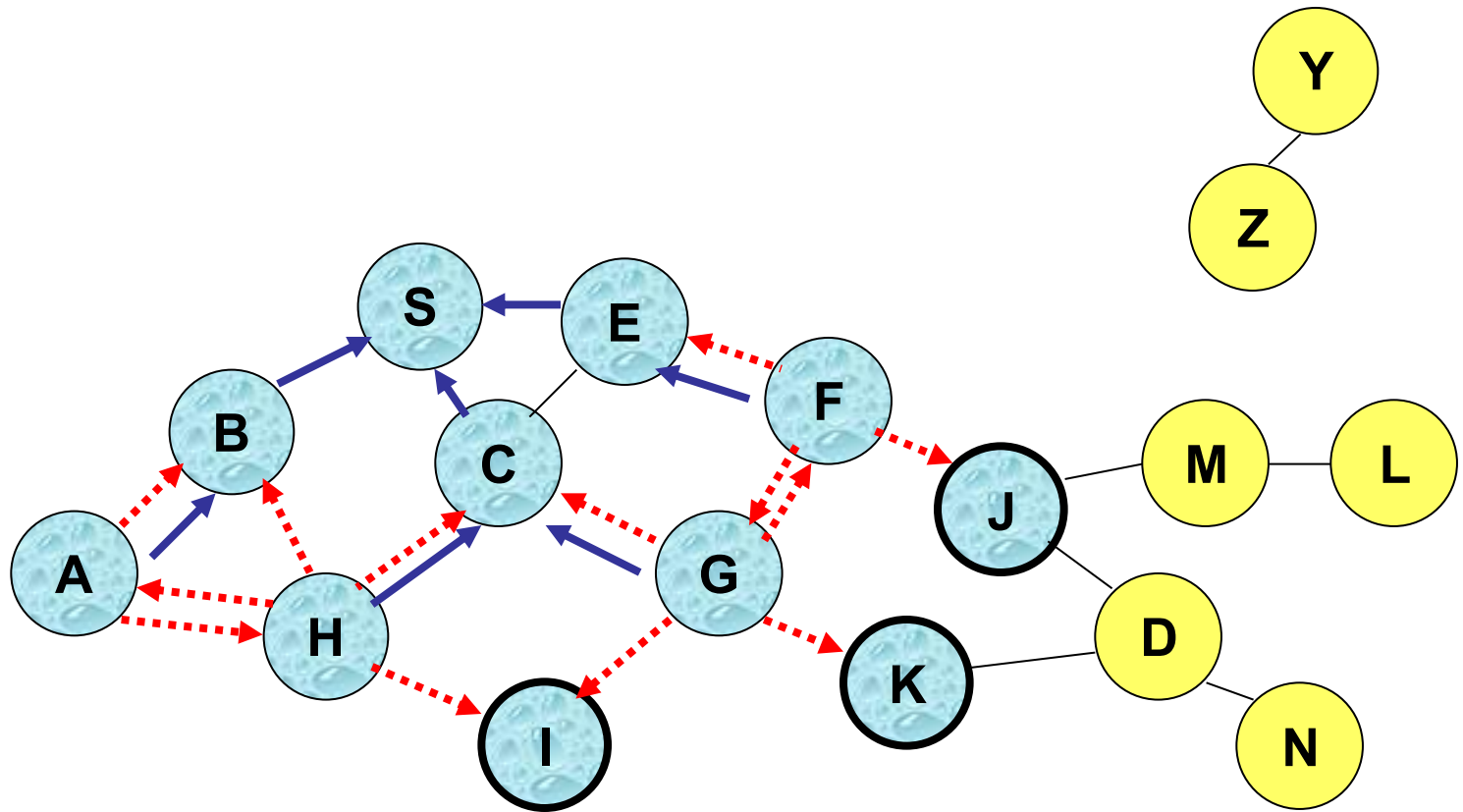
.....> Represents transmission of RREQ

Route Requests in AODV



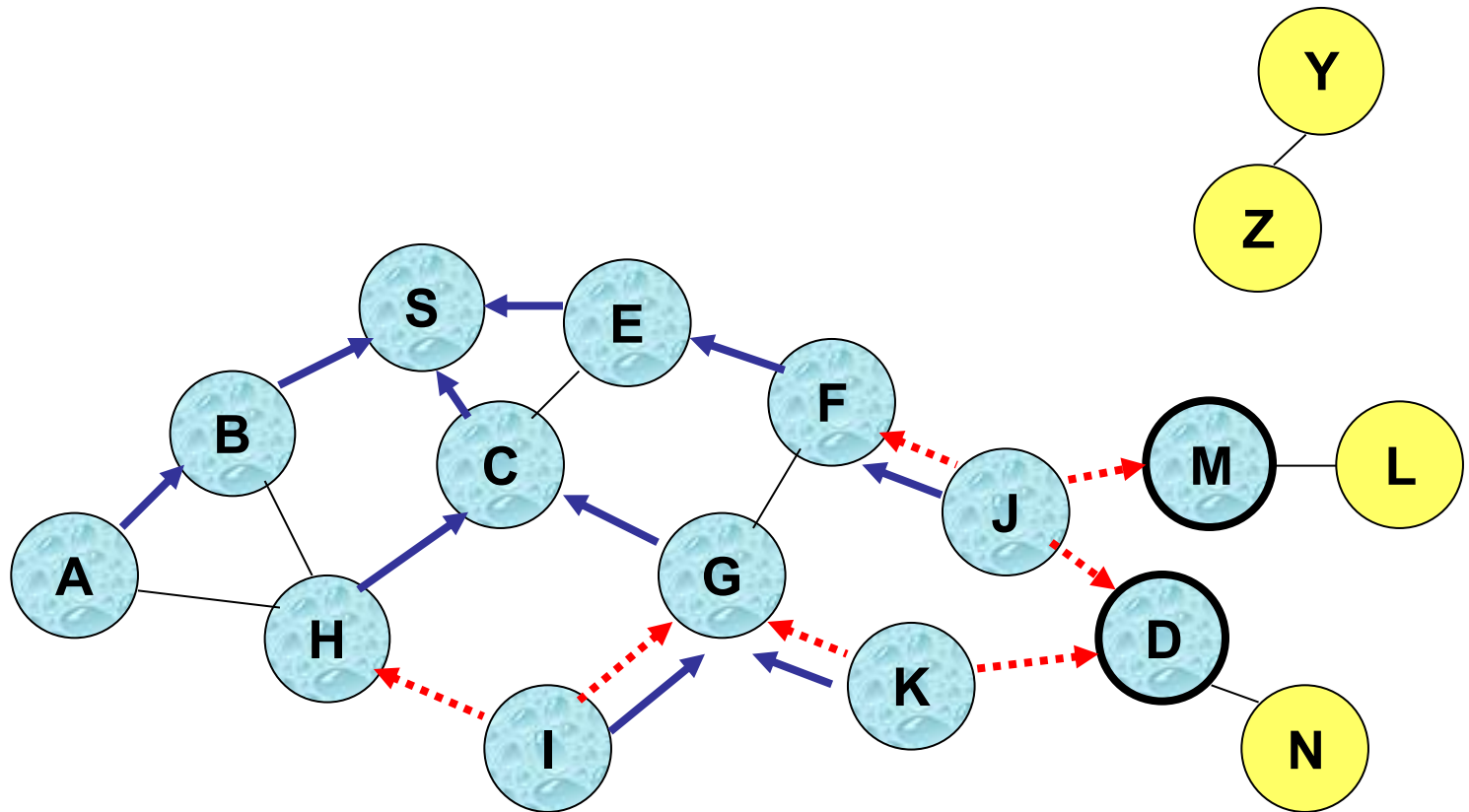
← Represents links on Reverse Path

Reverse Path Setup in AODV

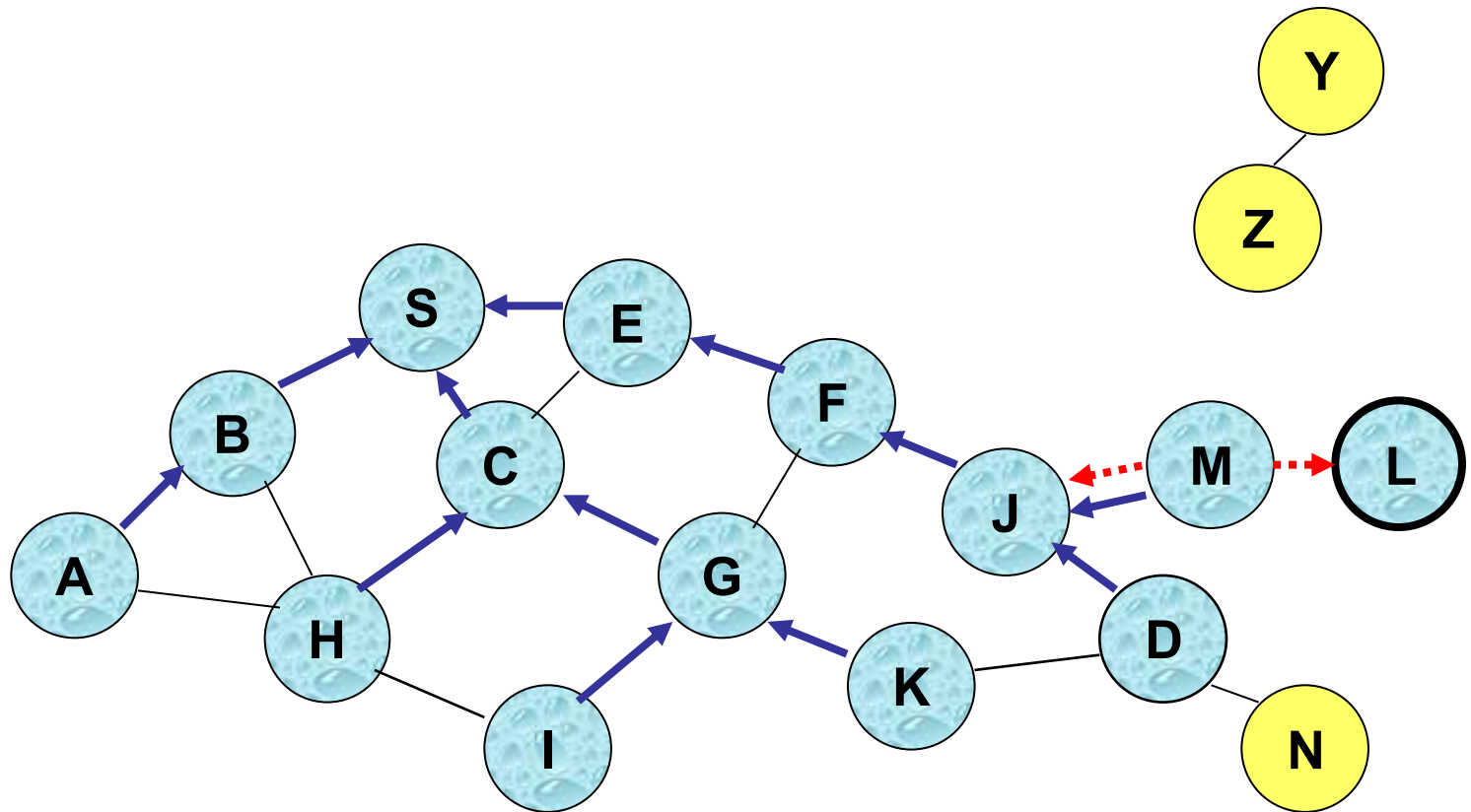


- Node C receives RREQ from G and H, but does not forward it again, because node C has **already forwarded RREQ** once

Reverse Path Setup in AODV

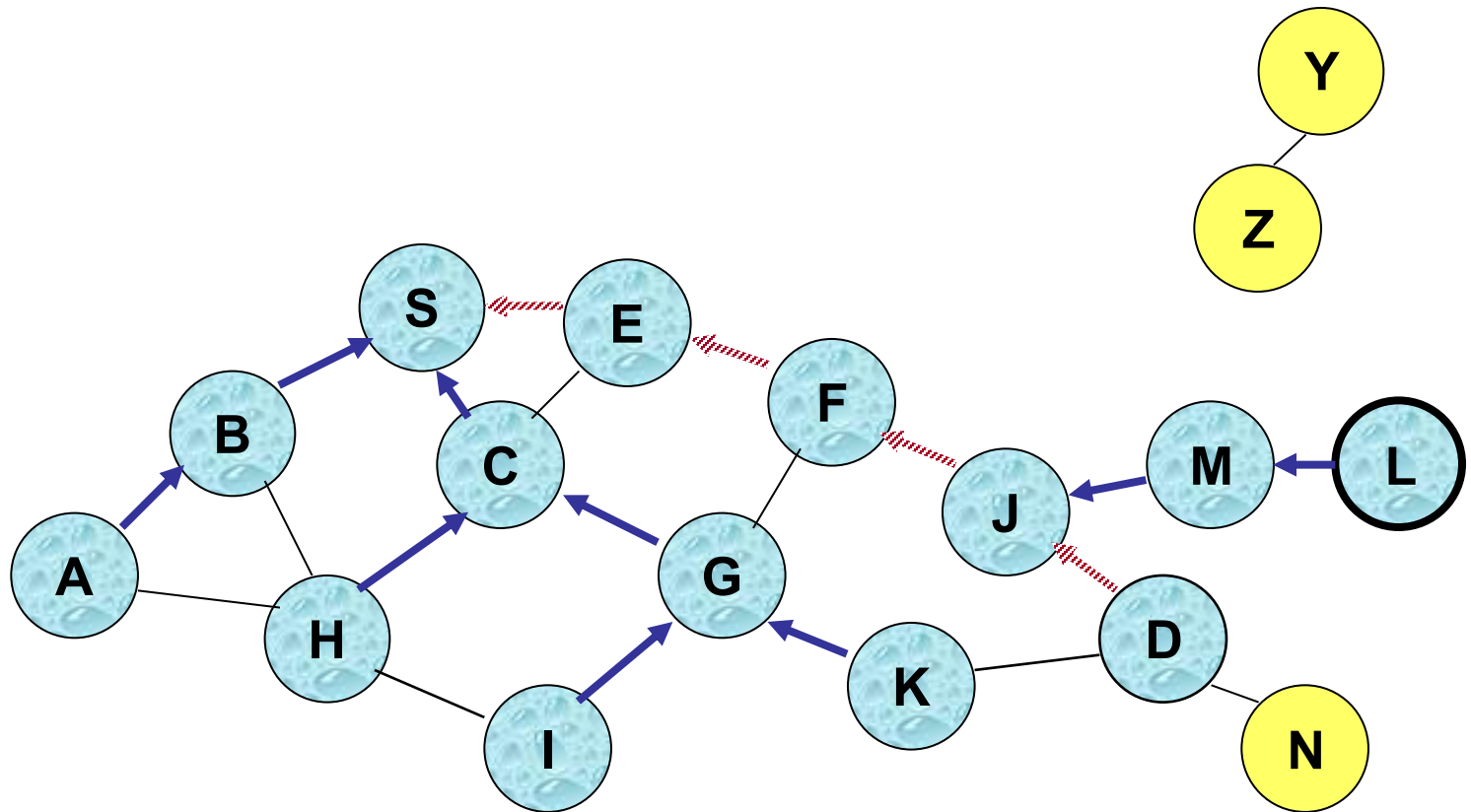


Reverse Path Setup in AODV



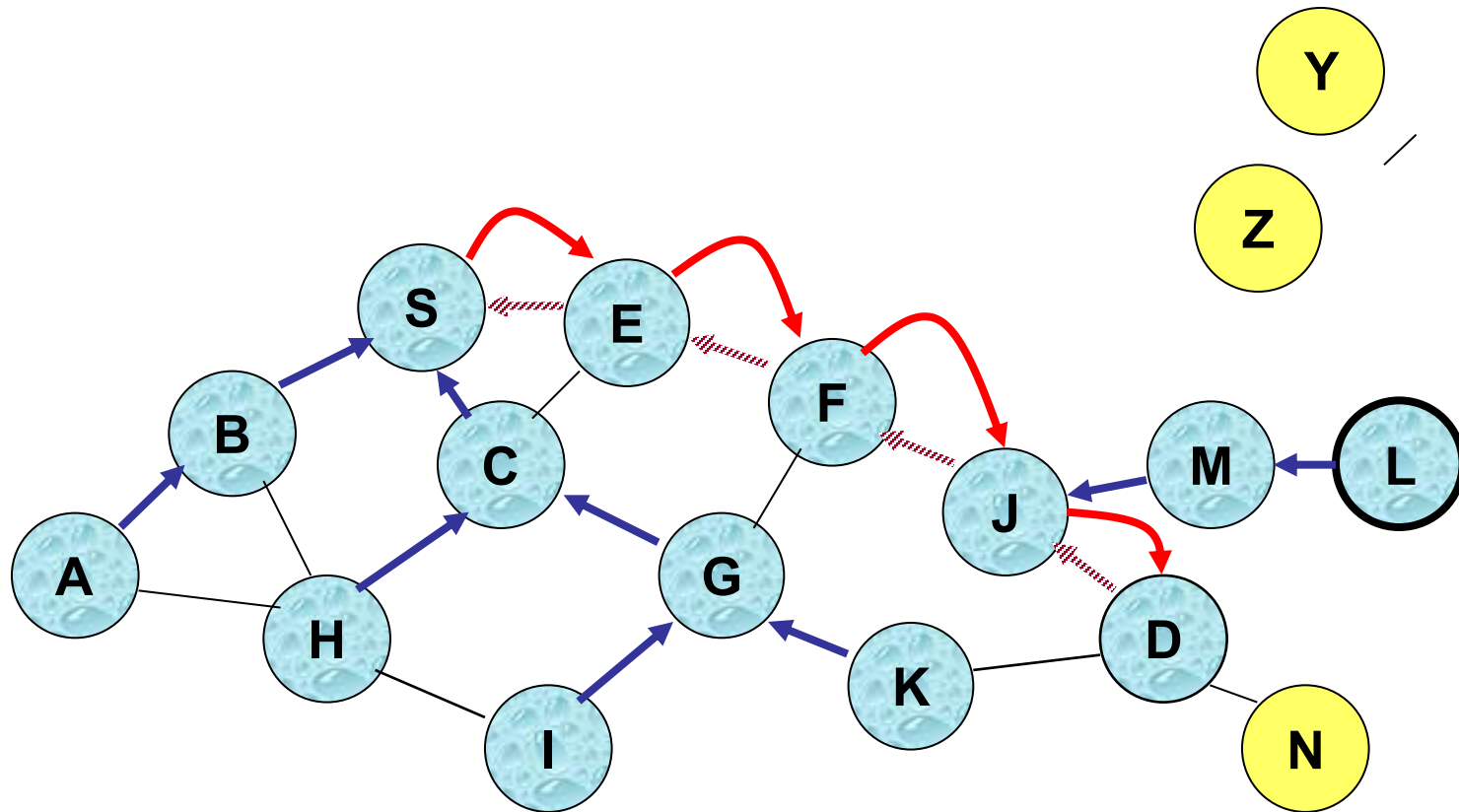
- Node D **does not forward** RREQ, because node D is the **intended target** of the RREQ

Route Reply in AODV



 Represents links on path taken by RREP

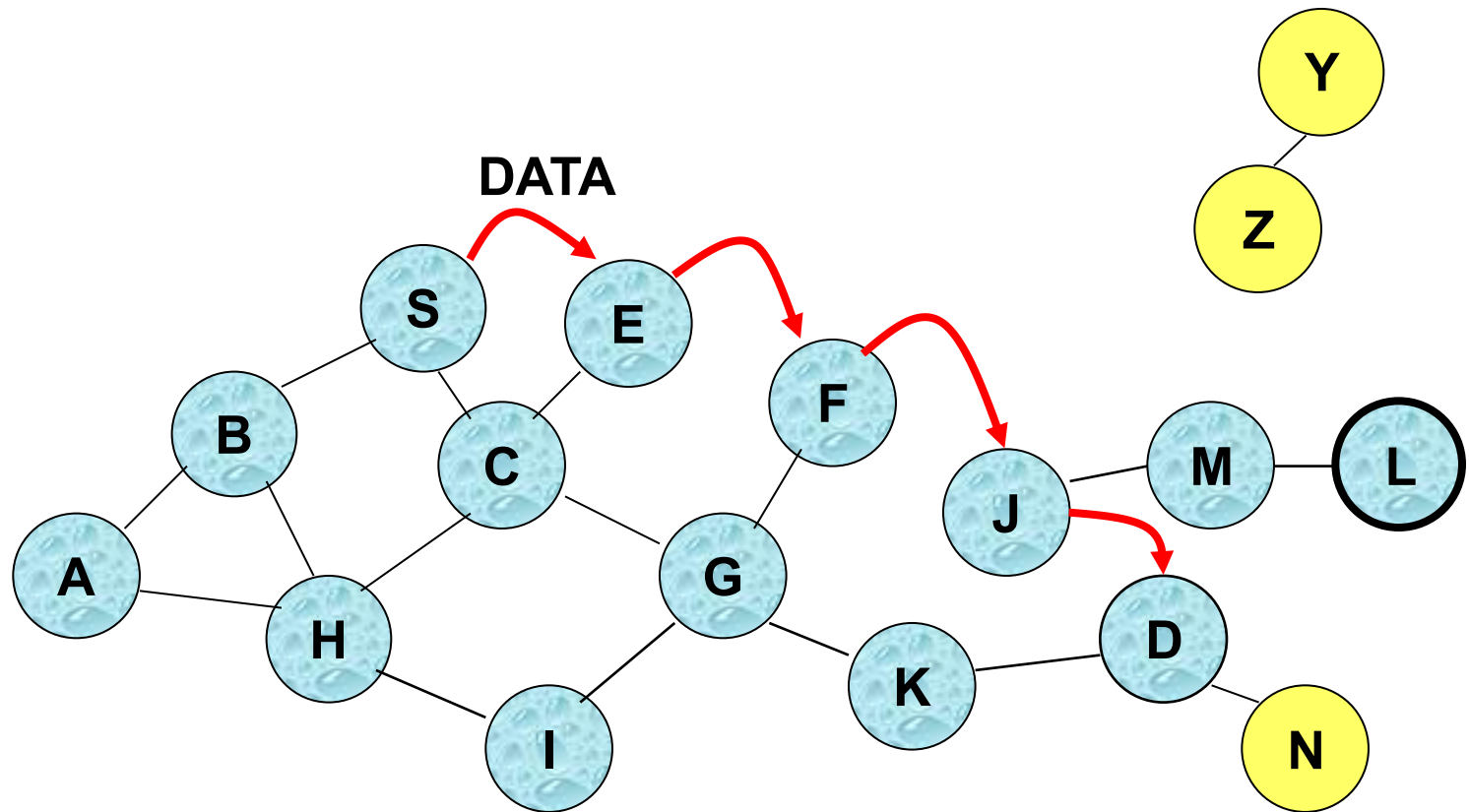
Forward Path Setup in AODV



Forward “next-hops” are setup when RREP travels along the reverse path – there is no actual path stored

 Represents a **next-hop** entry on the forward path to D

Data Delivery in AODV



Routing table entries used to forward data packet.

Route is *not* included in packet header.

Timeouts

- A routing table entry maintaining a reverse path is purged after a timeout interval
 - timeout should be long enough to allow RREP to come back
- A routing table entry maintaining a forward path is purged if *not used* for a *active_route_timeout* interval
 - if no data is being sent using a particular routing table entry, that entry will be deleted from the routing table (even if the route may actually still be valid)

Link Failure Detection

- *Hello* messages: Neighboring nodes periodically exchange hello message
- Absence of hello message is used as an indication of link failure
- Alternatively, failure to receive several MAC-level acknowledgement may be used as an indication of link failure

Destination Sequence Numbers in AODV

- An **intermediate node** (not the destination) may also send a **Route Reply (RREP)** provided that it knows a **more recent path** than the one previously known to sender S
- To determine whether the path known to an intermediate node is more recent, *destination sequence numbers* are used
- The likelihood that an intermediate node will send a Route Reply when using AODV not as high as DSR
 - A new Route Request by node S for a destination is assigned a higher destination sequence number. An intermediate node which knows a route, but with a smaller sequence number, **cannot send** Route Reply

Link Failure Reporting

- A neighbor of node X is considered **active** for a routing table entry if the neighbor sent a packet within *active_route_timeout* interval which was forwarded using that entry
- When the next hop link in a routing table entry breaks, all **active** neighbors are informed
- Link failures are propagated by means of Route Error messages, which also update destination sequence numbers

Route Error

- When node X is unable to forward packet P (from node S to node D) on link (X,Y), it generates a RERR message
- Node X increments the destination sequence number for D cached at node X
- The incremented sequence number N is included in the RERR
- When node S receives the RERR, it initiates a new route discovery for D using destination sequence number at least as large as N
- When node D receives the route request with destination sequence number N , node D will set its sequence number to N , unless it is already larger than N

Optimization: Expanding Ring Search [Perkins00]

- Route Requests are initially sent with small Time-to-Live (TTL) field, to limit their propagation
 - DSR also includes a similar optimization
- If no Route Reply is received, then larger TTL tried

Summary: AODV

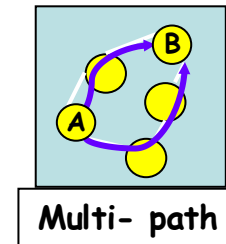
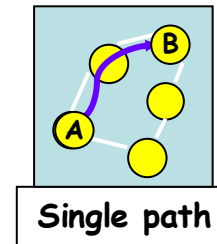
- Routes need not be included in packet headers
- Nodes maintain routing tables containing entries only for routes that are in active use
- At most one next-hop per destination maintained at each node
 - DSR may maintain several routes for a single destination
- Unused routes expire even if topology does not change

Hierarchical Routing

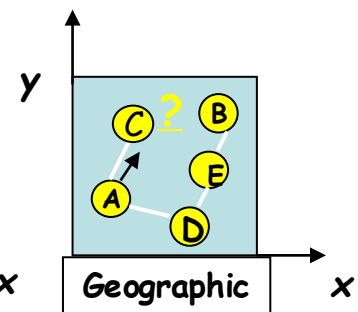
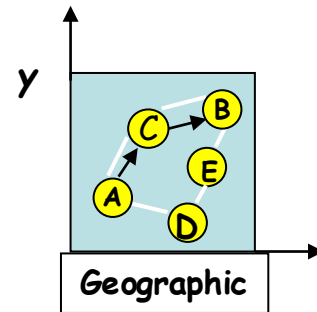
- In flat routing, all nodes had identical responsibilities
- Some schemes propose giving special responsibilities to a subset of nodes
 - Even if all nodes are physically identical
- Core-based schemes are examples of such schemes

Other Classes of Routing Protocols

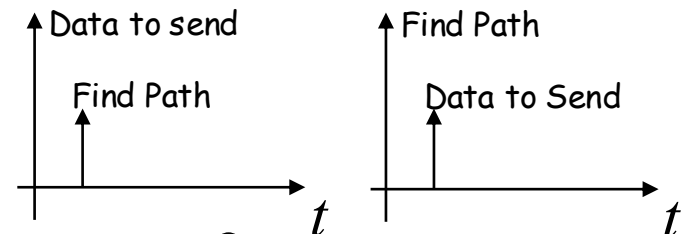
- **Single Path vs. Multi-Path**
 - Shortest delay
 - Largest available bandwidth
 - Largest residual power
 - Large expected lifetime



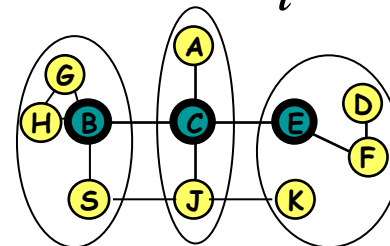
- **Unicast vs. Multicast**
- **Stateful vs. Geographic / “Stateless”**



- **On-Demand (Reactive) vs. Proactive**



- **Flat vs. Hierarchical**



DSR and AODV in NS-3

- Both DSR and AODV (and others) are implemented in NS-3 as open-source software
- Nice animations available using NetAnim tool
- Lab 4